GUMBEL FLUCTUATIONS FOR THE COVER TIME OF NEAR-COLLISIONS IN DYSON BROWNIAN MOTION

CÉSAR GALINDO

ABSTRACT. Let $\lambda(t)$ be a stationary Dyson Brownian motion, circular or of Ornstein–Uhlenbeck type, at inverse temperature $\beta \geq 1$. Let τ_i be the first time the i-th spacing falls below ε, and $\tau_{\mathrm{cov}} = \max_i \tau_i$ the cover time. We show that when near-collisions are rarer than the relaxation of the spectrum, the hitting times behave as independent exponentials: their normalized values converge to a Poisson process with intensity $e^{-x}dx$, and τ_{cov} has Gumbel fluctuations. The rate of the i-th spacing is $\Lambda_i = 2a_i/(\beta n\, I_\beta(\varepsilon))$, where a_i is the collision coefficient, $\mathbb{P}(g_i \leq x) \sim a_i x^{\beta+1}/(\beta+1)$, and $I_\beta(\varepsilon)$ equals $\varepsilon^{1-\beta}/(\beta-1)$ for $\beta > 1$ and $\log(1/\varepsilon)$ for $\beta = 1$. The results hold for the circular ensemble and in the bulk of the Hermite ensemble, and for initial laws with bounded L^2 density with respect to equilibrium. For the unitary circular ensemble they read $\mathbb{P}\big(\frac{(n^2-1)\varepsilon}{24\pi}\tau_{\mathrm{cov}} - \log n \leq x\big) \to e^{-e^{-x}}$ whenever $\varepsilon n^2 (\log n)^{7/2} \to 0$. The key estimate is an exact formula for the mean inverse square distance from a collision to the other particles, equal to $\beta(n^2-4)/(10(2\beta-1))$ for the circular β-ensemble, which we derive from loop equations of the circular Jacobi ensemble. When near-collisions are frequent but $n\varepsilon \to 0$, we show that each τ_i is of order $1/\Lambda_i$ and that $\tau_{\mathrm{cov}} = \Lambda_i^{-1} n^{o(1)}$, and we reduce the Gumbel law to a one-sided local law for collision coefficients of local equilibrium measures, which holds on average over the boundary condition. As an application, the cover time is the time at which every pair of neighbouring quasienergies of a white-noise driven quantum evolution has undergone an avoided crossing of width ε.

1. INTRODUCTION

1.1. The question.

In the normalization

$$d\lambda_k = \sqrt{\tfrac{2}{\beta n}}\, dB_k + \frac{1}{n}\sum_{j\neq k} \frac{dt}{\lambda_k - \lambda_j} - \frac{\lambda_k}{2}\, dt, \qquad k = 1, \dots, n, \tag{1}$$

Dyson Brownian motion is stationary under the Hermite β-ensemble, whose empirical measure approaches the semicircle law $\varrho(x) = \frac{1}{2\pi}\sqrt{4-x^2}$. The spacing $g_i = \lambda_{i+1} - \lambda_i$ is of order $1/(n\varrho)$, diffuses with coefficient $4/(\beta n)$, and is pushed away from zero by the drift $2/(ng_i)$. Near zero, ng_i behaves on the time scale $1/n$ as a Bessel process of dimension $\beta+1$, which for $\beta \geq 1$ never hits the origin.

Fix $\varepsilon > 0$ and an index set I, and write

$$\tau_i = \inf\{t \geq 0 :\ g_i(t) \leq \varepsilon\}, \qquad \tau_{\mathrm{cov}} = \max_{i\in I} \tau_i.$$

The cover time τ_{cov} is the first time at which every spacing in I has come within ε of a collision. If near-collisions are rare and weakly dependent, the spacings behave as the coupons of a coupon collector, and τ_{cov} should have Gumbel fluctuations around $\log|I|$ divided by the rate. We prove this, identify the rate, and describe the extremal process of the late hitting times.

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1.2. **The hazard.** Write p_i for the equilibrium density of g_i. The *collision coefficient*

$$a_i := \lim_{x\downarrow 0} x^{-\beta} p_i(x) \tag{2}$$

exists and is finite and positive (Lemma 2.3). Write

$$I_\beta(\varepsilon) = \frac{\varepsilon^{1-\beta}}{\beta-1} \quad (\beta>1), \qquad I_\beta(\varepsilon) = \log\frac{1}{\varepsilon} \quad (\beta=1),$$

so that $\int_\varepsilon^r g^{-\beta} dg = (1+o(1))I_\beta(\varepsilon)$ as $\varepsilon\to 0$, for any fixed $r>0$. The *hazard* of the i-th spacing is

$$\Lambda_i := \frac{2a_i}{\beta n\, I_\beta(\varepsilon)}, \tag{3}$$

the flux through the level ε of a Bessel process of dimension $\beta+1$ with diffusion coefficient $4/(\beta n)$ and stationary density $a_i x^\beta$ near the origin. For $\beta>1$ this is $\Lambda_i = \frac{2(\beta-1)}{\beta n} a_i \varepsilon^{\beta-1}$. In the bulk, $a_i \approx C_\beta(n\varrho(\gamma_i))^{\beta+1}$, where γ_i is the classical location and C_β the small-distance coefficient of the two-point function of Sine_β (Remark 5.3); thus $\Lambda_i \asymp n^\beta \varepsilon^{\beta-1}$ for $\beta>1$ and $\Lambda_i \asymp n/\log(1/\varepsilon)$ for $\beta=1$.

1.3. **Main results.** The circular Dyson process is

$$d\theta_k = \sqrt{\tfrac{2}{\beta n}}\, dB_k + \frac{1}{2n}\sum_{j\neq k} \cot\frac{\theta_k-\theta_j}{2}\, dt, \qquad k=1,\dots,n, \tag{4}$$

stationary under the circular β-ensemble. Its n cyclic spacings ($g_n = \theta_1 + 2\pi - \theta_n$) share one collision coefficient, a ratio of Morris integrals (Proposition 5.1):

$$a_\beta(n) = \frac{(n-1)\,\Gamma(1+\frac{\beta}{2})^n}{2\pi\,\Gamma(1+\frac{n\beta}{2})} \prod_{j=0}^{n-3} \frac{\Gamma(\frac{\beta}{2}j+2\beta+1)\,\Gamma(\frac{\beta}{2}(j+1)+1)}{\Gamma(\frac{\beta}{2}j+\beta+1)^2\,\Gamma(1+\frac{\beta}{2})}. \tag{5}$$

For $\beta=1,2,4$ it is an explicit polynomial in n, divided by 24, 24π and 1080π respectively (Corollary 5.2).

Theorem A (Circular ensemble)**.** *Let $\beta\geq 1$, start (4) from the circular β-ensemble, and let $\lambda_n = 2a_\beta(n)/(\beta n\, I_\beta(\varepsilon_n))$ satisfy $\lambda_n(\log n)^{(\beta+5)/2}\to 0$. Then, with $\tau_{\mathrm{cov}} = \max_{1\leq i\leq n}\tau_i$,*

- *(i)* $\mathbb{P}(\lambda_n\tau_{\mathrm{cov}} - \log n \leq x) \to \exp(-e^{-x})$ *for every $x\in\mathbb{R}$;*
- *(ii)* $\sum_{i=1}^n \delta_{\lambda_n\tau_i - \log n}$ *converges in distribution, on $(x_0,\infty]$ for every $x_0\in\mathbb{R}$, to the Poisson process with intensity $e^{-x}\,dx$;*
- *(iii)* $\sup_{t\geq 0}|\mathbb{P}(\lambda_n\tau_i > t) - e^{-t}| \to 0$.

In particular $\mathbb{P}(\lambda_n\tau_{\mathrm{cov}} - \log n\leq x)\to\exp(-e^{-x})$ in the three classical cases

$$\lambda_n = \frac{n^2-1}{12\,n\log(1/\varepsilon_n)}, \qquad \lambda_n = \frac{(n^2-1)\,\varepsilon_n}{24\pi}, \qquad \lambda_n = \frac{(n^2-1)(4n^2-1)\,\varepsilon_n^3}{720\pi}$$

for $\beta=1,2,4$ respectively, whenever

$$\frac{n(\log n)^3}{\log(1/\varepsilon_n)}\to 0, \qquad \varepsilon_n n^2(\log n)^{7/2}\to 0, \qquad \varepsilon_n^3 n^4(\log n)^{9/2}\to 0.$$

When the hazards differ the centering is implicit. For an index set I_n let $t_n(x)$ solve

$$\sum_{i\in I_n} e^{-\Lambda_i t} = e^{-x}, \qquad x > -\log|I_n|, \tag{6}$$

and let $x_n = t_n^{-1}$, with $x_n(0) = -\log|I_n|$.

Theorem B (Hermite ensemble). *Let $\beta \geq 1$, start (1) in equilibrium, let $0 < a < b < 1$ and $I_n \subset [an, bn]$ with $|I_n| \to \infty$, and assume that $\Lambda_{\max} = \max_{I_n} \Lambda_i$ satisfies $\Lambda_{\max}(\log n)^{(\beta+5)/2} \to 0$. Then $\sum_{i\in I_n} \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x}dx$; in particular $\mathbb{P}(\tau_{\mathrm{cov}} \leq t_n(x)) \to \exp(-e^{-x})$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \to 0$ uniformly in $i \in I_n$.*

When the collision coefficients are known to relative precision $o(1/\log n)$, Laplace's method makes t_n explicit, and the centering acquires a $\log\log n$ term (Proposition 8.1). For the bulk window $I_n = [n/4, 3n/4]$ of the Gaussian unitary ensemble, if $a_i = \frac{\pi^2}{3}(n\varrho(\gamma_i))^3(1+o(1/\log n))$, then

$$\mathbb{P}\Big(\Lambda_* \tau_{\mathrm{cov}} - \log n + \log\log n - \log c_* \leq x\Big) \to \exp(-e^{-x}),$$

where $\Lambda_* = \frac{\pi^2}{3}\varrho(\gamma_*)^3 n^2 \varepsilon$ is the smallest hazard, attained at the two ends of the window, $\gamma_* \approx -0.8079$ is the first quartile of the semicircle law, and $c_* = (4-\gamma_*^2)^{3/2}/(3\pi|\gamma_*|) \approx 0.8042$ (Proposition 8.2). The last spacings to be hit sit near the ends of the window, where the density is smallest; their number is of order $n/\log n$, not n. The assumption on a_i holds on average over indices (Section 11.2).

Theorem C (Warm starts). *The conclusions of Theorems A and B hold for the process started from any law $\nu_n = h_n\mu$ with $\sup_n \|h_n\|_{L^2(\mu)} < \infty$.*

For instance, the circular ensemble conditioned to have no eigenvalue in a fixed arc of length $2\pi s/n$, $0 < s < 1$, has $\|h_n\|^2 \leq (1-s)^{-1}$, since the expected number of eigenvalues in the arc is s.

Beyond the slow regime near-collisions are more frequent than the relaxation of the spectrum, and the global spectral gap is of no use. We call *intermediate* the regime $\Lambda_i \to \infty$, $n\varepsilon \to 0$. There we determine the order of the hitting times unconditionally, and reduce the Gumbel law to a static statement, Hypothesis 9.5: under the local equilibrium measure of a mesoscopic window with rigid exterior, the collision coefficient of a spacing is at least $(1 - o(1/\log n))a_i$, and the inverse square distances seen from a collision obey the bound of (P3) up to a factor $n^{o(1)}$. Both statements hold on average over the exterior.

Theorem D (Intermediate regime). *Let $\beta > 1$, consider the circular ensemble with any I_n, or the Hermite ensemble with $I_n \subset [an, bn]$, $0 < a < b < 1$, both in equilibrium, and assume $n\varepsilon_n \to 0$.*

- *(i) The laws of $\Lambda_i\tau_i$, $i \in I_n$, $n \geq 1$, form a tight family on $(0, \infty)$.*
- *(ii) $\log(\Lambda_{\min}\tau_{\mathrm{cov}})/\log n \to 0$ in probability, and for every $\delta > 0$ the expected number of $i \in I_n$ with $\tau_i > (1-\delta)\log|I_n|/\Lambda_{\max}$ tends to infinity.*
- *(iii) Assume $n\varepsilon_n \leq n^{-\delta}$ for some $\delta > 0$, and Hypothesis 9.5 with some $\xi < \delta(\beta-1)/(2(2\beta+1))$. Then $\mathbb{P}(\tau_{\mathrm{cov}} \leq t_n(x)) \to \exp(-e^{-x})$, every order statistic of the τ_i has the corresponding limit, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \to 0$.*

1.4. **Scope.** The condition $\Lambda_i \ll (\log n)^{-(\beta+5)/2}$ says that near-collisions are rarer than the relaxation of the whole spectrum, whose rate, the spectral gap, is a constant. For $\beta > 1$ it reads $\varepsilon \ll n^{-\beta/(\beta-1)}$ up to a power of $\log n$; at $\beta = 1$ it becomes $\log(1/\varepsilon) \gg n(\log n)^3$. The regime lies far below the static extremes: at a fixed time the smallest bulk spacing is of order $n^{-(\beta+2)/(\beta+1)}$ [6, 19], much larger than $n^{-\beta/(\beta-1)}$, so at a typical time no spacing is below ε, and the cover time is a dynamical extreme. Between $n^{-\beta/(\beta-1)}$ and n^{-1} lies the intermediate regime of Theorem D; there the infinite-volume dynamics has no spectral gap [30], and Poincaré inequalities on mesoscopic windows take its place.

1.5. **Method and related work.** Cover times of random walks have Gumbel fluctuations by Poisson approximation of the number of unvisited sites [1, 5, 14]; for random walks

on vertex-transitive graphs these fluctuations are characterized by a diameter condition [7]. For interacting particles the difficulty is the joint survival of several spacings. We avoid couplings with independent Bessel processes and work with the Dirichlet form of the n-particle dynamics.

(a) Survival of the spacings in S up to time t is avoidance of $A_S = \bigcup_{i\in S}\{g_i \le \varepsilon\}$. Its probability lies between $\pi(f^2)e^{-t\mathcal{E}(f)/\pi(f^2)}$, for test functions f vanishing on A_S, and $e^{-\lambda_{A_S}t}$, with λ_{A_S} the principal Dirichlet eigenvalue (Lemma 3.1). Matching the two to relative precision $o(1/\log n)$ gives the Gumbel law through factorial moments (Theorem 3.6). With a spectral gap the killed semigroup has a single eigenvalue below the gap, as in Aldous–Brown [2], and multi-time survival factorizes; this gives the point process (Theorem 3.7). The gap of Dyson Brownian motion follows from Brascamp–Lieb (Proposition 2.1).

(b) The gradients of non-adjacent spacings are orthogonal, so the energy of a function vanishing on A_S dominates the sum of its energies along these directions. Adjacent spacings meet at angle $2\pi/3$; the Gram matrix of a run of adjacent directions has norm below 2, which removes clusters (Proposition 4.2).

(c) Along the direction of one spacing the Gibbs density is $g^\beta\Phi(g,y)$ with Φ non-increasing in g (Lemma 2.2). This gives the exact upper bound $\mathbb{P}(g_i \le x) \le a_i x^{\beta+1}/(\beta+1)$. For the lower bound we compare a function vanishing on A_i with its mean on a slab $\{R \le g_i \le 2R\}$, under the Gibbs measure itself; the only input from the configuration is the mean of the inverse square distances seen from a collision. Three integrations by parts against the collision law, the first loop equations of a charged ensemble, compute this moment exactly (Section 6).

(d) In the intermediate regime the global gap is replaced by Brascamp–Lieb on windows of K particles with rigid exterior, whose Poincaré constant is of order n/K, and the eigenvalue bound by a decay estimate for $\|P_s^D\mathbf{1}_D\|^2$, which only involves functions with values in $[0,1]$ (Section 9). This is close in spirit to the local relaxation flow of Erdős, Schlein and Yau [17, 26].

Hitting times of rare sets in reversible chains are approximately exponential [2, 3]; capacities and principal Dirichlet eigenvalues of small sets are the tools of metastability [11]. Small spacings of random matrices at a fixed time are described in [6, 18, 19, 8], the last through the dynamics; high-probability lower bounds on gaps of Dyson Brownian motion uniformly over a time interval appear in [27]. Loop equations for circular ensembles are developed in [32]. Poisson point processes for extremes and factorial moments are classical [28, 23].

Outline. Section 2 sets up the Dyson processes and collision coefficients, and Section 3 proves the abstract limit theorems for rare hitting times of reversible diffusions. Section 4 computes the principal Dirichlet eigenvalue of unions of near-collision sets, assuming the moment bound of Condition 2.4. Section 5 computes the collision coefficients of the circular ensemble, and Section 6 proves Condition 2.4 through loop equations. Section 7 proves Theorems A, B and C, Section 8 treats variable density, and Section 9 the intermediate regime and Theorem D. Section 10 applies Theorem A to the quasienergies of white-noise driven quantum evolutions, and Section 11 collects remarks on clusters, on the collision coefficients of the Gaussian unitary ensemble, and on the questions that remain.

2. Dyson processes as reversible diffusions

2.1. **The two models.** *Hermite.* Write $W = \{\lambda \in \mathbb{R}^n : \lambda_1 < \cdots < \lambda_n\}$ and

$$\mu(d\lambda) = Z^{-1}e^{-V(\lambda)}\,d\lambda, \qquad V(\lambda) = \frac{\beta n}{4}\sum_k \lambda_k^2 - \beta\sum_{j<k}\log(\lambda_k - \lambda_j).$$

The generator of (1) is $Lf = \frac{1}{\beta n}(\Delta f - \nabla V \cdot \nabla f)$, with Dirichlet form

$$\mathcal{E}(f) = \frac{1}{\beta n}\int |\nabla f|^2 \, d\mu. \tag{7}$$

For $\beta \geq 1$, (1) has a unique strong solution which never leaves W [4, Prop. 4.3.5]; it is the μ-symmetric diffusion associated with the closure $(\mathcal{E}, \mathcal{F})$ of (7) on $C_c^\infty(W)$, and since ∂W is polar, $\mathcal{F}$ contains the bounded Lipschitz functions on W [22].

Circular. Let $\theta_1 < \cdots < \theta_n < \theta_1 + 2\pi$, $g_k = \theta_{k+1} - \theta_k$ for $k < n$ and $g_n = \theta_1 + 2\pi - \theta_n$. The spacings depend only on the projection of θ to $H = \{\sum_k \theta_k = 0\}$, and the projected process is a diffusion on the convex set $W^\circ = \{\theta \in H : \theta_1 < \cdots < \theta_n < \theta_1 + 2\pi\}$, reversible for

$$\mu(d\theta) = Z^{-1} e^{-V(\theta)} \, d\theta, \qquad V(\theta) = -\beta \sum_{j<k} \log\Big|2 \sin \frac{\theta_k - \theta_j}{2}\Big|,$$

with Lebesgue measure on H and Dirichlet form (7), gradients taken in H [13]. The spacings have the law of the circular β-ensemble, invariant under cyclic relabelling.

2.2. **Spectral gap.**

Proposition 2.1. *For $\beta \geq 1$, $n \geq 2$ and $f \in \mathcal{F}$, $\mathrm{Var}_\mu(f) \leq c_{\mathrm{P}}^{-1} \mathcal{E}(f)$, with $c_{\mathrm{P}} = \frac{1}{2}$ in the Hermite case and $c_{\mathrm{P}} = \frac{1}{4}$ in the circular case.*

Proof. In both cases μ has density e^{-V} on an open convex set, with V smooth there and tending to $+\infty$ at the boundary. By the Brascamp–Lieb inequality [12], $\nabla^2 V \geq \rho$ implies $\mathrm{Var}_\mu(f) \leq \rho^{-1} \int |\nabla f|^2 d\mu = \rho^{-1} \beta n \, \mathcal{E}(f)$. In the Hermite case $\nabla^2 V \geq \frac{\beta n}{2}$, since $-\log(\lambda_k - \lambda_j)$ is convex. In the circular case, for $v \in H$,

$$v^{\mathsf{T}} \nabla^2 V \, v = \frac{\beta}{4} \sum_{j<k} \frac{(v_k - v_j)^2}{\sin^2 \frac{\theta_k - \theta_j}{2}} \geq \frac{\beta}{4} \sum_{j<k} (v_k - v_j)^2 = \frac{\beta n}{4} |v|^2. \qquad \square$$

2.3. **Spacing coordinates and collision coefficients.** Fix a spacing i. Write $w_i = \frac{1}{2}(e_{i+1} - e_i)$ (with $e_{n+1} = e_1$ in the circular case), so that $\partial_{w_i} g_i = 1$ and $|w_i|^2 = \frac{1}{2}$. We write a configuration as (g, y) with $g = g_i$: in the Hermite case $y = (\lambda_1, \ldots, \lambda_{i-1}, m, \lambda_{i+2}, \ldots, \lambda_n)$, $m = \frac{1}{2}(\lambda_i + \lambda_{i+1})$, so that $\lambda_i = m - \frac{g}{2}$ and $\lambda_{i+1} = m + \frac{g}{2}$, with Jacobian one; in the circular case the same holds in an affine chart of H, the constant Jacobian being absorbed in Z. Then $\partial_g = \partial_{w_i}$ and

$$e^{-V} = g^\beta \, \Phi_i(g, y), \tag{8}$$

where, with j ranging over the indices other than $i, i+1$ and up to factors free of g,

$$\Phi_i(g, y) = e^{-\frac{\beta n}{8} g^2} \prod_j \Big|(m - \lambda_j)^2 - \tfrac{g^2}{4}\Big|^\beta$$

in the Hermite case and

$$\Phi_i(g, y) = \Big(\frac{\sin(g/2)}{g/2}\Big)^\beta \prod_j \Big|4 \sin^2 \tfrac{m - \theta_j}{2} - 4 \sin^2 \tfrac{g}{4}\Big|^\beta$$

in the circular case. The admissible g are $0 < g < g_{\max}(y)$, where $g_{\max}(y)/2$ is the distance from m to the nearest other particle. Every factor of $\Phi_i(\cdot, y)$ is nonincreasing on $(0, g_{\max}(y))$. Write

$$D_i(y) = \sum_j (m - \lambda_j)^{-2} \quad \text{or} \quad D_i(y) = \sum_j \Big(2 \sin \frac{m - \theta_j}{2}\Big)^{-2}.$$

Lemma 2.2. *For admissible (g, y), $\Phi_i(g, y) \leq \Phi_i(0, y)$. If moreover $g^2 D_i(y) \leq 2$, then (g, y) is admissible and $\Phi_i(g, y) \geq \Phi_i(0, y) \exp(-\frac{\beta g^2}{2}(D_i(y) + n))$.*

Proof. In the Hermite case $\Phi_i(g,y)/\Phi_i(0,y) = e^{-\beta n g^2/8}\prod_j (1-u_j)^\beta$ with $u_j = g^2/(4(m-\lambda_j)^2)$, which is at most one for admissible g. If $g^2 D_i \le 2$, then $u_j \le \frac12$ for every j, g is admissible, and $\log(1-u) \ge -2u$ gives the lower bound. In the circular case the ratio is $(\frac{\sin(g/2)}{g/2})^\beta \prod_j (1-u_j)^\beta$ with $u_j = \sin^2\frac{g}{4}/\sin^2\frac{m-\theta_j}{2}$; now $2\sum_j u_j \le \frac{g^2}{2} D_i$, and $\frac{\sin x}{x} \ge e^{-x^2/3}$ for $0 \le x \le 1$. □

Lemma 2.3. *Write $c_i(y) = Z^{-1}\Phi_i(0,y)$ for y with m strictly between its neighbours, and $a_i = \int c_i\,dy$. Then $0 < a_i < \infty$, $p_i(x) \le a_i x^\beta$ for all $x > 0$, and $x^{-\beta}p_i(x) \to a_i$ as $x \downarrow 0$. In particular $\mathbb{P}(g_i \le x) \le a_i x^{\beta+1}/(\beta+1)$.*

Proof. Up to a constant, c_i is the density of an $(n-1)$-particle Gibbs measure with a charge 2β at m, hence integrable. By (8), $p_i(x) = x^\beta \int Z^{-1}\Phi_i(x,y)\mathbf{1}\{x < g_{\max}(y)\}\,dy$; the integrand is at most $c_i(y)$ and tends to it as $x \downarrow 0$. □

The *collision law* $\hat\nu_i(dy) = a_i^{-1}c_i(y)\,dy$ is the law of the other particles given that λ_i and λ_{i+1} have merged. The capacity estimates of Section 4 rest on the following condition, which we prove in Section 6 for the circular ensemble and for bulk windows of the Hermite ensemble (Proposition 6.4).

Definition 2.4. An index set I_n satisfies *Condition (P)* if there is $C_0 \ge 1$ such that for all large n and all $i \in I_n$:

(P1) $\max_{I_n} a_i \le C_0 \min_{I_n} a_i$;
(P2) $C_0^{-1} n^{\beta+1} \le a_i \le C_0 n^{\beta+1}$;
(P3) $\int D_i\,d\hat\nu_i \le C_0 n^2$.

3. Rare hitting times in reversible diffusions

Throughout this section X is a diffusion on E, symmetric with respect to a probability measure π, with regular Dirichlet form $(\mathcal{E},\mathcal{F})$, started from π unless stated otherwise. For a closed set A write $T_A = \inf\{t \ge 0 : X_t \in A\}$, $D = E \setminus A$, $\mathcal{F}_A = \{f \in \mathcal{F} : \tilde f = 0 \text{ q.e. on } A\}$, and

$$\lambda_A = \inf\{\mathcal{E}(f)/\pi(f^2) :\ f \in \mathcal{F}_A,\ f \ne 0\}.$$

The part of X on D is associated with $(\mathcal{E},\mathcal{F}_A)$, and $\mathbb{P}_\pi(T_A > t) = \langle \mathbf{1}_D, P_t^D \mathbf{1}_D\rangle_\pi$ with P^D the killed semigroup [22, Thms. 4.4.2–4.4.3].

3.1. Survival estimates.

Lemma 3.1. *For every closed A, every $t \ge 0$ and every $f \in \mathcal{F}_A$ with $0 \le f \le 1$,*

$$\pi(f^2)\exp\Big(-t\,\frac{\mathcal{E}(f)}{\pi(f^2)}\Big) \le \mathbb{P}_\pi(T_A > t) \le e^{-\lambda_A t}.$$

Moreover $\mathbb{E}_\pi T_A \le 1/\lambda_A$ and $\mathbb{E}_\pi T_A \ge \pi(f)^2/\mathcal{E}(f)$.

Proof. Write ν for the spectral measure of $\mathbf{1}_D$ for the killed generator; it is carried by $[\lambda_A,\infty)$ and has mass $\pi(D) \le 1$, hence $\mathbb{P}_\pi(T_A > t) = \int e^{-st}\nu(ds) \le e^{-\lambda_A t}$ and $\mathbb{E}_\pi T_A \le 1/\lambda_A$. Since P_t^D preserves positivity and $0 \le f \le \mathbf{1}_D$, $\langle \mathbf{1}_D, P_t^D \mathbf{1}_D\rangle \ge \langle f, P_t^D f\rangle = \int e^{-st}\nu_f(ds)$, where ν_f has mass $\pi(f^2)$ and first moment $\mathcal{E}(f)$; Jensen's inequality gives the lower bound. Finally $u = \int_0^\infty P_t^D \mathbf{1}_D\,dt$ satisfies $\mathcal{E}(u) = \langle u, \mathbf{1}_D\rangle = \mathbb{E}_\pi T_A$ and $\pi(f) = \mathcal{E}(f,u) \le \mathcal{E}(f)^{1/2}\mathcal{E}(u)^{1/2}$. □

Lemma 3.2. *Let A be closed and suppose $\mathcal{E}(f) \ge a\,\pi(f^2) - b$ for every $f \in \mathcal{F}_A$ with $0 \le f \le 1$, where $a > 0$ and $b \ge 0$. Then $\mathbb{P}_\pi(T_A > t) \le e^{-at} + b/a$ for all $t \ge 0$.*

Proof. Write $u_s = P_s^D \mathbf{1}_D$ and $\phi(s) = \|u_s\|^2$. For $s > 0$, $u_s \in \mathcal{F}_A$ and $0 \le u_s \le 1$, and $\phi'(s) = -2\mathcal{E}(u_s) \le -2a\phi(s) + 2b$; thus $\phi(s) \le e^{-2as} + b/a$. Since P^D is self-adjoint, $\mathbb{P}_\pi(T_A > t) = \langle \mathbf{1}_D, P_t^D \mathbf{1}_D\rangle = \phi(t/2)$. □

Lemma 3.3. *Assume* $\mathrm{Var}_\pi(f) \le c_{\mathrm{P}}^{-1}\mathcal{E}(f)$ *on* $\mathcal{F}$, *and let* $0 < \lambda_A < c_{\mathrm{P}}$. *Then* λ_A *is a simple isolated eigenvalue of the killed generator with eigenfunction* $\varphi_A \ge 0$, $\|\varphi_A\| = 1$, *the rest of the spectrum lies in* $[c_{\mathrm{P}}, \infty)$, *and with* $c_A = \langle \mathbf{1}_D, \varphi_A\rangle^2$,

$$0 \le \mathbb{P}_\pi(T_A > t) - c_A e^{-\lambda_A t} \le (1 - c_A)e^{-c_{\mathrm{P}} t}, \qquad c_A \ge \lambda_A\big(\mathbb{E}_\pi T_A - c_{\mathrm{P}}^{-1}\big).$$

Proof. Every two-dimensional subspace of $\mathcal{F}_A$ contains $f \ne 0$ with $\pi(f) = 0$, for which $\mathcal{E}(f) \ge c_{\mathrm{P}}\pi(f^2)$; by min–max the spectral projection on $[0, c_{\mathrm{P}})$ has rank at most one. If φ is an eigenfunction, $|\varphi| \in \mathcal{F}_A$ has the same Rayleigh quotient, hence $\varphi = \pm|\varphi|$. With ν as above, $\nu(\{\lambda_A\}) = c_A$ and $\nu([c_{\mathrm{P}}, \infty)) \le 1 - c_A$; this gives the first bound, and integrating in t gives $\mathbb{E}_\pi T_A \le c_A/\lambda_A + c_{\mathrm{P}}^{-1}$. □

Lemma 3.4. *In the setting of Lemma 3.3, let* $A_1 \supseteq \cdots \supseteq A_k$ *be closed with* $0 < \lambda_{A_l} < c_{\mathrm{P}}$, $0 = t_0 < t_1 < \cdots < t_k$, *and* $\Delta_l = t_l - t_{l-1}$. *Then*

$$\mathbb{P}_\pi\big(X_s \notin A_l \text{ for } s \le t_l,\ l \le k\big) = \Theta \prod_l e^{-\lambda_{A_l}\Delta_l} + \rho,$$

with $1 - 5\sum_l(1 - c_{A_l}) \le \Theta \le 1$ *and* $|\rho| \le 2^k e^{-c_{\mathrm{P}} \min_l \Delta_l}$.

Proof. Write $Q_l f = P^{D_l}_{\Delta_l}(f\mathbf{1}_{D_l})$. By the Markov property the probability is $\langle 1, Q_1 \cdots Q_k 1\rangle$. By Lemma 3.3, $Q_l = e^{-\lambda_{A_l}\Delta_l}\varphi_l\langle\varphi_l, \cdot\rangle + R_l$ with $\|R_l\| \le e^{-c_{\mathrm{P}}\Delta_l}$. The term without R_l's is $\Theta\prod_l e^{-\lambda_{A_l}\Delta_l}$, $\Theta = \langle 1, \varphi_1\rangle\langle\varphi_1, \varphi_2\rangle \cdots \langle\varphi_k, 1\rangle$, and each of the other $2^k - 1$ terms is at most $e^{-c_{\mathrm{P}} \min \Delta_l}$. All inner products lie in $[0, 1]$; $\|\varphi_l - 1\|^2 \le 2(1 - c_{A_l})$ gives $\langle\varphi_l, \varphi_{l+1}\rangle \ge 1 - 2(1 - c_{A_l}) - 2(1 - c_{A_{l+1}})$, and $\langle 1, \varphi_l\rangle \ge c_{A_l}$. □

Lemma 3.5. *In the setting of Lemma 3.3, let* $\nu = h\pi$ *with* $h \in L^2(\pi)$. *Then* $\mathbb{P}_\nu(T_A > t) \le \|h\|e^{-\lambda_A t}$ *and*

$$\big|\mathbb{P}_\nu(T_A > t) - c_A e^{-\lambda_A t}\big| \le \|h - 1\|(1 - c_A)^{1/2}e^{-\lambda_A t} + \|h\|(1 - c_A)^{1/2}e^{-c_{\mathrm{P}} t}.$$

In the setting of Lemma 3.4, the same probability under $\mathbb{P}_\nu$ *equals* $\Theta_\nu\prod_l e^{-\lambda_{A_l}\Delta_l} + \rho_\nu$ *with* $|\Theta_\nu - \Theta| \le \|h - 1\|(1 - c_{A_1})^{1/2}$ *and* $|\rho_\nu| \le 2^k\|h\|e^{-c_{\mathrm{P}} \min_l \Delta_l}$.

Proof. We have $\mathbb{P}_\nu(T_A > t) = \langle h, P^D_t\mathbf{1}_D\rangle$ and $\|P^D_t\mathbf{1}_D\| \le e^{-\lambda_A t}$, which gives the first bound. Write $\mathbf{1}_D = \sqrt{c_A}\,\varphi_A + r$ with $r \perp \varphi_A$ and $\|r\|^2 = \pi(D) - c_A \le 1 - c_A$. By Lemma 3.3, $\|P^D_t r\| \le e^{-c_{\mathrm{P}} t}\|r\|$, and

$$\langle h, P^D_t\mathbf{1}_D\rangle = \sqrt{c_A}\,e^{-\lambda_A t}\langle h, \varphi_A\rangle + \langle h, P^D_t r\rangle.$$

Since $\pi(h - 1) = 0$ and $\langle 1, \varphi_A\rangle = \sqrt{c_A}$, we have $\langle h, \varphi_A\rangle - \sqrt{c_A} = \langle h - 1, \varphi_A - \sqrt{c_A}\rangle$, and $\|\varphi_A - \sqrt{c_A}\|^2 = 1 - c_A$; this gives the second bound. The multi-time statement follows from the proof of Lemma 3.4, with $\langle 1, \varphi_1\rangle$ replaced by $\langle h, \varphi_1\rangle$ and $\|Q_l\| \le 1$. □

3.2. **Limit theorems.** Consider now a sequence of such diffusions with finite index sets I_n, $N_n = |I_n| \to \infty$, a graph on I_n of degree at most d, closed sets A_i and rates $\Lambda_i > 0$. For $S \subset I_n$ write $A_S = \bigcup_{i\in S} A_i$, $\lambda_S = \lambda_{A_S}$ and $\Lambda_S = \sum_{i\in S}\Lambda_i$; S is *separated* if it contains no two neighbours. Write $\tau_i = T_{A_i}$, $\Lambda_{\max} = \max\Lambda_i$, $\Lambda_{\min} = \min\Lambda_i$, and t_n, x_n as in (6). We consider the conditions:

(A) $\Lambda_{\max} \le C_0\Lambda_{\min}$;
(B) for every m and D there are $\eta_n = \eta_n(m)$ with $\eta_n \log N_n \to 0$ and C_D such that for all S with $|S| \le m$ and all $t \ge 0$,

$$\text{(9)}\quad \mathbb{P}_\pi(T_{A_S} > t) \le \exp\Big(-(1-\eta_n)t\sum_C w_C\Big) + C_D N_n^{-D}, \qquad w_C = \Big(1 + \tfrac{1}{3}\mathbf{1}_{\{|C|\ge 2\}}\Big)\min_{i\in C}\Lambda_i,$$

the sum over the connected components C of S. By Lemma 3.1, (B) holds with $C_D = 0$ if

$$\lambda_S \geq (1-\eta_n) \sum_C w_C; \tag{10}$$

(C) for every separated S with $|S| \leq m$ there is $f_S \in \mathcal{F}_{A_S}$, $0 \leq f_S \leq 1$, with $\pi(f_S^2) \geq 1-\eta_n$ and $\mathcal{E}(f_S) \leq (1+\eta_n)\Lambda_S\, \pi(f_S^2)$.

Theorem 3.6. *Under (A)-(C), for every $x \in \mathbb{R}$ the number of i with $\tau_i > t_n(x)$ converges in law to a Poisson variable with mean e^{-x}. Hence $\mathbb{P}(\max_i \tau_i \leq t_n(x)) \to \exp(-e^{-x})$, the k-th largest $\tau_{(k)}$ satisfies $\mathbb{P}(\tau_{(k)} \leq t_n(x)) \to e^{-e^{-x}} \sum_{j<k} e^{-jx}/j!$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \leq 3\eta_n(1)$ for all i.*

Proof. By Lemma 3.1, (B) and (C), for separated S with $|S| \leq m$,

$$(1-\eta_n)e^{-(1+\eta_n)\Lambda_S t} \leq \mathbb{P}(T_{A_S} > t) \leq e^{-(1-\eta_n)\Lambda_S t} + C_D N_n^{-D}. \tag{11}$$

For $S = \{i\}$ this gives the last claim, the additive term being absorbed for large n. Since $N_n e^{-\Lambda_{\max} t} \leq \sum_i e^{-\Lambda_i t} \leq N_n e^{-\Lambda_{\min} t}$,

$$(\log N_n + x)/\Lambda_{\max} \leq t_n(x) \leq (\log N_n + x)/\Lambda_{\min}. \tag{12}$$

Fix x, $t = t_n(x)$, $W = \#\{i : \tau_i > t\}$ and $m \geq 1$. Then $\mathbb{E}(W)_m = \sum \mathbb{P}(T_{A_S} > t)$ over ordered m-tuples of distinct indices, S their set. There are at most N_n^m tuples; with $D = m+1$ the additive terms in (9) contribute $o(1)$ and may be dropped. For separated S, (11) and $\Lambda_S t \leq mC_0(\log N_n + x)$ give $\mathbb{P}(T_{A_S} > t) = (1+o(1)) \prod_{i\in S} e^{-\Lambda_i t}$ uniformly. Summed over all tuples, with repetitions allowed, the product equals $(\sum_i e^{-\Lambda_i t})^m = e^{-mx}$; tuples with a repeated index or two neighbours contribute to this sum at most $m^2(d+1)e^{-\Lambda_{\min} t} e^{-(m-1)x} \to 0$, since $\Lambda_{\min} t \geq (\log N_n + x)/C_0$.

For non-separated S, (9) gives $\mathbb{P}(T_{A_S} > t) \leq \prod_C e^{-(1-\eta_n)w_C t}$ up to the dropped term. We group the tuples by the shape of their decomposition into components; the sum over placements factorizes over components. A singleton contributes $\sum_i e^{-(1-\eta_n)\Lambda_i t} \leq e^{\eta_n \Lambda_{\max} t} e^{-x} = O(1)$. A component of size $s \geq 2$ is a connected set containing its minimizing index, and there are at most $(ed)^s$ such sets per vertex; thus it contributes at most $(ed)^s e^{-x} e^{\eta_n \Lambda_{\max} t} \max_i e^{-(\frac{1}{3} - \frac{4}{3}\eta_n)\Lambda_i t} \to 0$. Hence $\mathbb{E}(W)_m \to e^{-mx}$ for every m, and W converges to a Poisson variable with mean e^{-x}. The statements on order statistics follow from $\{\tau_{(k)} \leq t\} = \{W < k\}$. □

Theorem 3.7. *Assume (A), (C), the eigenvalue bound (10), $\mathrm{Var}_\pi(f) \leq c_{\mathrm{P}}^{-1}\mathcal{E}(f)$ with $c_{\mathrm{P}} \geq c > 0$ for all n, and $\Lambda_{\max} \log N_n \to 0$. Then for every $x_0 \in \mathbb{R}$, $\sum_i \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x}dx$.*

Proof. For separated S with $|S| \leq m$, (C) and Lemma 3.1 give $\lambda_S \leq (1+\eta_n)\Lambda_S$ and, since $\pi(f_S) \geq \pi(f_S^2)$, $\mathbb{E}_\pi T_{A_S} \geq (1-\eta_n)/((1+\eta_n)\Lambda_S)$; with (10), $\lambda_S < c$ for large n, and Lemma 3.3 gives $c_{A_S} \geq 1 - \eta_n'$ with $\eta_n' \to 0$.

Fix $x_0 < a_1 < b_1 \leq a_2 < \cdots < b_k \leq \infty$, $J_l = (t_n(a_l), t_n(b_l)]$ and $M_l = \#\{i : \tau_i \in J_l\}$. We show $\mathbb{E}\prod_l (M_l)_{m_l} \to \prod_l (e^{-a_l} - e^{-b_l})^{m_l}$; by the method of moments the M_l then converge jointly to independent Poisson variables, which characterizes convergence to the Poisson process [23, Ch. 4]. The expectation is a sum over families of distinct indices of $\mathbb{P}(\tau_i \in J_{l(i)},\ i \in S)$. Since $t_n' \geq 1/\Lambda_{\max}$, distinct endpoints satisfy $|t_n(a) - t_n(b)| \geq |a-b|/\Lambda_{\max}$, and $t_n(a_1) \geq (\log N_n + a_1)/\Lambda_{\max}$.

For separated S we expand $\mathbf{1}\{\tau \in J_l\} = \mathbf{1}\{\tau > t_n(a_l)\} - \mathbf{1}\{\tau > t_n(b_l)\}$. Each resulting term $\mathbb{P}(\tau_i > u_i,\ i \in S)$ is the probability of avoiding a decreasing sequence of sets A_{S_p} on $[0, s_p]$, $s_1 < s_2 < \ldots$ the distinct values of the u_i. By Lemma 3.4 it equals $(1 + o(1)) \exp(-\sum_i \Lambda_i u_i) + O(2^m e^{-ch/\Lambda_{\max}})$ for some $h > 0$, because $\lambda_{S_p} = (1 + O(\eta_n))\Lambda_{S_p}$

and $\sum_i \Lambda_i u_i = O(\log N_n)$. Summing back, the main term is $\prod_i (e^{-\Lambda_i t_n(a_{l(i)})} - e^{-\Lambda_i t_n(b_{l(i)})})$, whose sum over all families with repetitions is $\prod_l (e^{-a_l} - e^{-b_l})^{m_l}$ by (6). Non-distinct or non-separated families, the $o(1)$ terms, and the terms $N_n^m 2^m e^{-ch/\Lambda_{\max}}$, which vanish because $\Lambda_{\max} \log N_n \to 0$, are negligible exactly as in the proof of Theorem 3.6, with $t = t_n(a_1)$. □

4. Capacity of spacing events

Throughout this section we work with the Dyson processes, with $A_i = \{g_i \le \varepsilon\}$, i and $i \pm 1$ neighbours (indices mod n in the circular case), and hazards (3). Write $I_\beta(\varepsilon, R) = \int_\varepsilon^R g^{-\beta} dg$ for $0 < \varepsilon < R$. Constants C depend only on β, C_0 and m.

Proposition 4.1. *Let S be separated, $|S| \le m$, $0 < \varepsilon < R \le 1$, and $f = \prod_{i \in S}(1 - h(g_i))$ with $h = 1$ on $[0, \varepsilon]$, $h(g) = I_\beta(g, R)/I_\beta(\varepsilon, R)$ on $[\varepsilon, R]$, and $h = 0$ on $[R, \infty)$. Then $f \in \mathcal{F}_{A_S}$, $0 \le f \le 1$,*

$$\mathcal{E}(f) \le \Lambda_S \frac{I_\beta(\varepsilon)}{I_\beta(\varepsilon, R)}, \qquad \pi(f^2) \ge 1 - \sum_{i \in S} \frac{a_i R^{\beta+1}}{\beta + 1}.$$

Proof. f is bounded, Lipschitz and vanishes on A_S. Since $\nabla g_i = 2w_i$ and $w_i \perp w_j$ for non-neighbours, $|\nabla f|^2 \le 2 \sum_{i \in S} h'(g_i)^2$. By Lemma 2.3, $\int h'(g_i)^2 d\mu \le a_i \int_\varepsilon^R h'(g)^2 g^\beta dg = a_i / I_\beta(\varepsilon, R)$, and $\frac{2}{\beta n} a_i / I_\beta(\varepsilon, R) = \Lambda_i I_\beta(\varepsilon)/I_\beta(\varepsilon, R)$. Finally $f = 1$ unless some $g_i < R$, and $\mu(g_i < R) \le a_i R^{\beta+1}/(\beta+1)$. □

The ratio in Proposition 4.1 is

$$\frac{I_\beta(\varepsilon, R)}{I_\beta(\varepsilon)} = 1 - \Big(\frac{\varepsilon}{R}\Big)^{\beta-1} \quad (\beta > 1), \qquad \frac{I_\beta(\varepsilon, R)}{I_\beta(\varepsilon)} = 1 - \frac{\log(1/R)}{\log(1/\varepsilon)} \quad (\beta = 1).$$

For a run C of s consecutive indices write $\mu_C = 1 + \cos \frac{\pi}{s+1}$, and $\mathcal{C}(S)$ for the decomposition of S into runs.

Proposition 4.2. *Assume Condition 2.4. Let $0 < nR \le \frac{1}{2}$ and $\varepsilon < R$, and write $\theta = (\varepsilon/R)^{\min(\beta-1,1)}$ for $\beta > 1$ and $\theta = 1/\log(1/\varepsilon)$ for $\beta = 1$. For every $S \subset I_n$ with $|S| \le m$,*

$$\lambda_S \ge (1 - \eta) \sum_{C \in \mathcal{C}(S)} \mu_C^{-1} \sum_{i \in C} \Lambda_i, \qquad \eta = C\Big((nR)^2 + \Big(\frac{\Lambda_{\max}}{(nR)^{\beta+1}}\Big)^{1/2} + \theta + \Lambda_{\max}\Big).$$

Proof. Geometry. Write $u_i = \sqrt{2} w_i$. For $|i - j| \ge 2$, $u_i \perp u_j$, and $\langle u_i, u_{i+1} \rangle = -\frac{1}{2}$; the Gram matrix of a run C is tridiagonal with largest eigenvalue μ_C, and different runs span orthogonal subspaces. Hence $|v|^2 \ge \sum_C \mu_C^{-1} \sum_{i \in C} \langle u_i, v \rangle^2$, and with $\langle u_i, \nabla f \rangle^2 = 2(\partial_{g_i} f)^2$,

$$\mathcal{E}(f) \ge \sum_C \mu_C^{-1} \sum_{i \in C} \mathcal{E}_i(f), \qquad \mathcal{E}_i(f) = \frac{2}{\beta n} \int (\partial_{g_i} f)^2 d\mu. \tag{13}$$

One direction. Let $f \in \mathcal{F}_{A_S}$. If $\pi(f) = 0$, then $\mathcal{E}(f) \ge c_{\mathrm{P}} \pi(f^2)$ and there is nothing to prove; if $\mathcal{E}(f) > 2\Lambda_S \pi(f)^2$, we go to the conclusion. Otherwise fix $i \in S$, write $w(g) = Z^{-1} g^\beta \Phi_i(g, y)$, $x(g) = \frac{\beta}{2} g^2 (D_i(y) + n)$, and $G = \{y : 2R^2 D_i(y) \le 1\}$. For $y \in G$ and $g \le 2R$ we have $g^2 D_i \le 2$ and $x(g) \le 2\beta$, because $nR^2 \le \frac{1}{4n}$; Lemma 2.2 gives $w(g) \ge c_i(y) g^\beta e^{-x(g)}$, and $(2R, y)$ is admissible. Thus, with $e^x - 1 \le e^{2\beta} x$ and $D_i + n \le R^{-2}$,

$$\int_\varepsilon^{2R} \frac{dg}{w(g)} \le \frac{1}{c_i(y)} \Big(I_\beta(\varepsilon) + C R^{-2} \int_\varepsilon^{2R} g^{2-\beta} dg \Big) \le \frac{(1 + C\theta)\, I_\beta(\varepsilon)}{c_i(y)};$$

indeed $R^{-2}\int_\varepsilon^{2R} g^{2-\beta}dg/I_\beta(\varepsilon)$ is at most $C/\log(1/\varepsilon)$ for $\beta=1$, $C(\varepsilon/R)^{\beta-1}$ for $1<\beta<3$, $C(\varepsilon/R)^2\log(2R/\varepsilon)$ for $\beta=3$ and $C(\varepsilon/R)^2$ for $\beta>3$, all bounded by $C\theta$. For almost every $y\in G$, $g\mapsto f(g,y)$ is absolutely continuous and vanishes at ε, and for $r\in[R,2R]$ the Cauchy–Schwarz inequality gives

$$f(r,y)^2\le\int_\varepsilon^{2R}\frac{dg}{w(g)}\int_\varepsilon^{2R}(\partial_g f)^2 w\,dg\le\frac{(1+C\theta)I_\beta(\varepsilon)}{c_i(y)}\int_\varepsilon^{2R}(\partial_g f)^2 w\,dg.$$

We multiply by $w(r)$ and integrate over $r\in[R,2R]$. Since $\Phi_i(r,y)\le\Phi_i(0,y)$, $\int_R^{2R}w\le c_i(y)M_R$ with $M_R=\int_R^{2R}r^\beta dr\ge R^{\beta+1}$. Integrating over $y\in G$ and using $\frac{2}{\beta n I_\beta(\varepsilon)}=\frac{\Lambda_i}{a_i}$,

$$\mathcal{E}_i(f)\ge\frac{\Lambda_i}{(1+C\theta)\,a_i M_R}\int_B f^2\,d\mu,\qquad B=\{g_i\in[R,2R]\}\cap\{y\in G\}.$$

The slab B carries almost the mass a_iM_R. Indeed $e^{-x}\ge1-x$ and $x(r)\le2\beta R^2(D_i+n)$ for $r\le2R$ give

$$\begin{aligned}\mu(B)&\ge M_R\int_G c_i(y)\big(1-2\beta R^2(D_i(y)+n)\big)dy\\&\ge a_iM_R\Big(1-\hat\nu_i(G^c)-2\beta R^2\Big(\int D_i\,d\hat\nu_i+n\Big)\Big)\ge a_iM_R\big(1-C(nR)^2\big),\end{aligned}$$

because $\hat\nu_i(G^c)\le2R^2\int D_i\,d\hat\nu_i\le2C_0(nR)^2$ by (P3). By Proposition 2.1, $\|f-\pi(f)\|^2\le\mathcal{E}(f)/c_{\mathrm P}\le2\Lambda_S\pi(f)^2/c_{\mathrm P}$, and

$$\int_B f^2\,d\mu\ge\Big(|\pi(f)|\mu(B)^{1/2}-\|f-\pi(f)\|\Big)^2\ge\mu(B)\,\pi(f)^2\big(1-\sqrt{\eta_2}\big)^2,$$

where

$$\eta_2=\frac{2\Lambda_S}{c_{\mathrm P}\,\mu(B)}\le\frac{C\Lambda_{\max}}{(nR)^{\beta+1}},$$

using $\Lambda_S\le m\Lambda_{\max}$ and (P2). Thus $\mathcal{E}_i(f)\ge(1-C\theta-C(nR)^2-2\sqrt{\eta_2})\Lambda_i\pi(f)^2$.

Conclusion. By (13), $\mathcal{E}(f)\ge(1-\eta')\Lambda_S^{\mathrm{eff}}\pi(f)^2$ with $\Lambda_S^{\mathrm{eff}}=\sum_C\mu_C^{-1}\sum_{i\in C}\Lambda_i\le\Lambda_S$, in all cases. Since $\pi(f^2)\le\pi(f)^2+c_{\mathrm P}^{-1}\mathcal{E}(f)$, $\mathcal{E}(f)/\pi(f^2)\ge(1-\eta'-4m\Lambda_{\max})\Lambda_S^{\mathrm{eff}}$. □

The proof uses the Gibbs measure itself on the slab B, not its approximation $c_i(y)g^\beta$; the moment (P3) enters only through the mass of B and the probability of G^c, both of order $(nR)^2$.

Corollary 4.3. *Assume Condition 2.4 and $\Lambda_{\max}(\log n)^{(\beta+5)/2}\to0$. Then conditions (A), (C) and the eigenvalue bound (10) of Section 3 hold for the sets A_i, $i\in I_n$, with $\eta_n\log n\to0$.*

Proof. (A) is (P1). For runs of length $s\ge2$, $s/\mu_C\ge\frac43$, and Proposition 4.2 gives (10) once $\eta\log n\to0$. Write $L=\log n$, $\kappa_n=\Lambda_{\max}L^{(\beta+5)/2}\to0$, $\omega=\min(\kappa_n^{-1/(\beta+1)},L)\to\infty$, and take $nR=(L\omega)^{-1/2}$. Then $(nR)^2L=\omega^{-1}\to0$ and

$$L\Big(\frac{\Lambda_{\max}}{(nR)^{\beta+1}}\Big)^{1/2}=\big(\Lambda_{\max}L^{(\beta+5)/2}\omega^{(\beta+1)/2}\big)^{1/2}\le\kappa_n^{1/4}\to0.$$

By (P2), $I_\beta(\varepsilon)=2a_i/(\beta n\Lambda_i)\ge cn^\beta/\Lambda_{\max}$. For $\beta>1$ this gives $(\varepsilon/R)^{\beta-1}=R^{1-\beta}/((\beta-1)I_\beta(\varepsilon))\le C\Lambda_{\max}(nR)^{1-\beta}/n\le CL^\beta/n$, in particular $\varepsilon<R$; for $\beta=1$ it gives $\theta\le C\Lambda_{\max}/n$. In both cases $\theta L\to0$, and $\eta L\to0$. For (C) we use Proposition 4.1 with the same R: $\sum_{i\in S}a_iR^{\beta+1}\le mC_0(L\omega)^{-(\beta+1)/2}=o(1/L)$, and $1-I_\beta(\varepsilon,R)/I_\beta(\varepsilon)$ equals $(\varepsilon/R)^{\beta-1}$ for $\beta>1$ and $\log(1/R)/\log(1/\varepsilon)\le C\Lambda_{\max}\log n/n$ for $\beta=1$, which is $O(n^{-1/2})$. □

5. The circular ensemble

Proposition 5.1. *The collision coefficient of the circular β-ensemble is* (5).

Proof. Write $P = Z_n^{-1}\prod_{j<k}|e^{i\theta_j}-e^{i\theta_k}|^\beta$ on $[0,2\pi)^n$, where $Z_n = (2\pi)^n\Gamma(1+\frac{n\beta}{2})/\Gamma(1+\frac{\beta}{2})^n$ [21]. All factors of P are bounded, and by dominated convergence and rotation invariance

$$\lim_{x\downarrow 0} x^{-\beta}\rho_2(\phi,\phi+x) = \frac{n(n-1)(2\pi)^{n-2}}{Z_n} M_{n-2}\big(\beta,\beta,\tfrac{\beta}{2}\big),$$

where, by Morris' identity [21],

$$\begin{aligned} M_N(a,b,\lambda) &= \int_{[-\pi,\pi]^N} \prod_j e^{\frac{i}{2}(a-b)\theta_j}|1+e^{i\theta_j}|^{a+b}\prod_{j<k}|e^{i\theta_j}-e^{i\theta_k}|^{2\lambda}\frac{d\theta}{(2\pi)^N} \\ &= \prod_{j=0}^{N-1}\frac{\Gamma(\lambda j+a+b+1)\,\Gamma(\lambda(j+1)+1)}{\Gamma(\lambda j+a+1)\,\Gamma(\lambda j+b+1)\,\Gamma(1+\lambda)}. \end{aligned}$$

For small x, pairs at distance x are consecutive up to $O(x^{2\beta+1})$, and the n spacings are identically distributed; thus $np(x) = 2\pi\rho_2(0,x) + O(x^{2\beta+1})$, which gives (5). □

Corollary 5.2. *For the orthogonal, unitary and symplectic circular ensembles,*

$$a_1(n) = \frac{n^2-1}{24}, \qquad a_2(n) = \frac{n(n^2-1)}{24\pi}, \qquad a_4(n) = \frac{n(n^2-1)(4n^2-1)}{1080\pi}.$$

Proof. For $\beta\in\{1,2,4\}$ the ratio $a_\beta(n+1)/a_\beta(n)$ read off from (5) involves only Gamma functions whose arguments differ by integers, and $\Gamma(x+1)=x\Gamma(x)$ reduces it to $\frac{n(n+2)}{(n-1)(n+1)}$, $\frac{n+2}{n-1}$ and $\frac{(n+2)(2n+3)}{(n-1)(2n-1)}$ respectively; these are the ratios of the right-hand sides. At $n=2$ both sides equal $\frac{1}{8}$, $\frac{1}{4\pi}$ and $\frac{1}{12\pi}$. □

With (3) the hazards of the three ensembles are $\frac{n^2-1}{12n\log(1/\varepsilon)}$, $\frac{(n^2-1)\varepsilon}{24\pi}$ and $\frac{(n^2-1)(4n^2-1)\varepsilon^3}{720\pi}$, the values in Theorem A.

Remark 5.3. By Stirling's formula, $(2\pi)^{\beta+1}n^{-\beta-1}a_\beta(n)\to C_\beta$, where

$$C_\beta = (\pi\beta)^\beta\frac{\Gamma(1+\frac{\beta}{2})^3}{\Gamma(1+\beta)\Gamma(1+\frac{3\beta}{2})}$$

is the small-distance coefficient of the two-point function of Sine_β [20, Ch. 13]: $C_1 = \pi^2/6$, $C_2 = \pi^2/3$ and $C_4 = 16\pi^4/135$, in agreement with Corollary 5.2. Hence $\lambda_n \sim \frac{2(\beta-1)}{\beta}C_\beta(2\pi)^{-\beta-1}n^\beta\varepsilon^{\beta-1}$ for $\beta>1$, and $\lambda_n\sim n/(12\log(1/\varepsilon))$ for $\beta=1$.

6. The configuration seen from a collision

In this section we compute the moment in (P3) exactly, by integrating by parts against the collision law. The three identities we use are the first loop equations of the charged ensemble that describes the other particles once two of them have merged; for the circular ensemble this is a circular Jacobi ensemble [20, Ch. 3], [32].

6.1. **The circular ensemble.** As a law of $n-2$ unordered points, the collision law of the circular ensemble is the law Q_m with density proportional to $\prod_j|e^{im}-e^{i\theta_j}|^{2\beta}\prod_{j<k}|e^{i\theta_j}-e^{i\theta_k}|^\beta$, averaged over a uniform m.

Proposition 6.1. *For the circular β-ensemble with $\beta\ge 1$ and every spacing i,*

$$\int D_i\,d\hat{\nu}_i = \frac{\beta\,(n^2-4)}{10(2\beta-1)}.$$

Proof. By rotation invariance we may take $m = 0$. Write $N = n-2$, p for the density of Q_0 on $(0,2\pi)^N$, $\mathbb{E}$ for the expectation under Q_0, and $c_j = \cot\frac{\theta_j}{2}$, $S_1 = \sum_j c_j$, $S_2 = \sum_j c_j^2$. Since $(2\sin\frac{\theta}{2})^{-2} = \frac{1}{4}(1+\cot^2\frac{\theta}{2})$, we must compute $\mathbb{E}D = \frac{1}{4}(N + \mathbb{E}S_2)$. We have

$$\partial_j \log p = \beta c_j + \frac{\beta}{2}\sum_{k\neq j} \cot\frac{\theta_j - \theta_k}{2}.$$

For a vector field V that is smooth off the sets $\{\theta_j = 0\}$ and $\{\theta_j = \theta_k\}$, bounded near the second and $O(|\theta_j|^{-1})$ near the first, $\int (\operatorname{div} V + V\cdot\nabla \log p)\, p = 0$. Indeed p vanishes to order $2\beta > 1$ on $\{\theta_j = 0\}$ and to order β on $\{\theta_j = \theta_k\}$, so the boundary terms vanish, and the integrands are $O(|\theta_j|^{2\beta-2})$ and $O(|\theta_j - \theta_k|^{\beta-1})$ there, hence integrable. We use two such fields.

For $V_j = c_j$ we have $\operatorname{div} V = -\frac{1}{2}\sum_j (1 + c_j^2) = -\frac{1}{2}(N + S_2)$. By the identity $(\cot x - \cot y)\cot(x-y) = -(1 + \cot x \cot y)$, symmetrization of the pair term gives

$$\sum_{j\neq k} c_j \cot\frac{\theta_j - \theta_k}{2} = \frac{1}{2}\sum_{j\neq k}(c_j - c_k)\cot\frac{\theta_j - \theta_k}{2} = -\frac{1}{2}\big(N(N-1) + S_1^2 - S_2\big),$$

and $V\cdot\nabla\log p = \beta S_2 - \frac{\beta}{4}(N(N-1) + S_1^2 - S_2)$.

For $V_j = S_1$ for every j, again $\operatorname{div} V = -\frac{1}{2}(N+S_2)$, and $V\cdot\nabla\log p = \beta S_1^2$, the pair term being antisymmetric in (j,k). The second field gives $\beta\,\mathbb{E}S_1^2 = \frac{1}{2}(N + \mathbb{E}S_2)$, and inserting this in the first,

$$\Big(\frac{5\beta}{4} - \frac{1}{2}\Big)\mathbb{E}S_2 = \frac{\beta}{4}N(N-1) + \frac{N}{2} + \frac{1}{8}(N + \mathbb{E}S_2), \qquad \mathbb{E}S_2 = \frac{2\beta N(N-1) + 5N}{5(2\beta - 1)}.$$

Hence $\mathbb{E}D = \frac{\beta N(N+4)}{10(2\beta-1)}$, which is the claim with $N = n-2$. □

In units where the mean spacing is one, Proposition 6.1 reads $(2\pi/n)^2 \int D_i\, d\hat\nu_i \to \frac{2\pi^2\beta}{5(2\beta-1)}$. At $\beta = 2$ the limit is also read off from a determinantal kernel.

Remark 6.2. The circular unitary ensemble is determinantal with kernel $K(x,y) = \langle v_y, v_x\rangle$, $v_x = (2\pi)^{-1/2}(e^{-ikx})_{k<n}$, and $\rho_3(p,q,z)/\rho_2(p,q) = \operatorname{dist}(v_z, \operatorname{span}\{v_p, v_q\})^2$. Letting $p, q \to m$, the one-point density of Q_m is $\rho^{(m)}(z) = \operatorname{dist}(v_z, \operatorname{span}\{v_m, \partial v_m\})^2$, which satisfies $\rho^{(m)} \le \frac{n}{2\pi}$ and $\rho^{(m)}(m+u) \le \frac{n^5u^4}{40\pi}$. After the scaling $u = 2\pi x/n$ it converges to $1 - s(x)^2 - \frac{3}{\pi^2}s'(x)^2$, $s(x) = \frac{\sin\pi x}{\pi x}$, the one-point function of the sine process conditioned on a double point at 0. The two bounds dominate $\rho^{(m)}(z)(2\sin\frac{u}{2})^{-2}$ after scaling, and Proposition 6.1 with $\beta = 2$ yields

$$\int_{\mathbb{R}} \frac{1 - s(x)^2 - \frac{3}{\pi^2}s'(x)^2}{x^2}\, dx = \frac{4\pi^2}{15}.$$

6.2. **The Hermite ensemble.** In the Hermite case the collision law $\hat\nu_i$ lives on $W_i = \{y = (\lambda_1, \dots, \lambda_{i-1}, m, \lambda_{i+2}, \dots, \lambda_n)$ increasing$\}$, with density

$$c_i(y) \propto e^{-\frac{\beta n}{2}m^2} \prod_j |m - \lambda_j|^{2\beta}\, e^{-\frac{\beta n}{4}\sum_j \lambda_j^2} \prod_{j<k} |\lambda_j - \lambda_k|^\beta,$$

where j,k range over the $n-2$ indices other than $i, i+1$; the factor $e^{-\beta n m^2/2}$ is the limit of $e^{-\beta n(\lambda_i^2 + \lambda_{i+1}^2)/4}$.

Proposition 6.3. *For the Hermite β-ensemble with $\beta \ge 1$ and every $1 \le i \le n-1$,*

$$\int D_i\, d\hat\nu_i = \frac{n\big(4\beta(n-2) + 1 - \beta n \int m^2\, d\hat\nu_i\big)}{10(2\beta - 1)} \le \frac{2\beta n^2}{5(2\beta - 1)}.$$

Proof. Write $N = n-2$, $T = \sum_j (m-\lambda_j)^{-1}$, and $\mathbb{E}$ for the expectation under $\hat{\nu}_i$. The density vanishes on ∂W_i, to order 2β on $\{m = \lambda_j\}$, and the fields below have singularities of order one there; integration by parts holds as in the proof of Proposition 6.1.

The field $T\,\partial_m$, with $\partial_m T = -D_i$ and $\partial_m \log c_i = -\beta n m + 2\beta T$, gives $2\beta\,\mathbb{E}T^2 = \mathbb{E}D_i + \beta n\,\mathbb{E}[mT]$. The field $m\,\partial_m$ gives $1 - \beta n \mathbb{E}m^2 + 2\beta\mathbb{E}[mT] = 0$. The field $\sum_j (\lambda_j - m)^{-1}\partial_{\lambda_j}$ has divergence $-D_i$; since

$$\partial_{\lambda_j} \log c_i = \frac{2\beta}{\lambda_j - m} - \frac{\beta n}{2}\lambda_j + \beta \sum_{k\neq j} \frac{1}{\lambda_j - \lambda_k}, \qquad \sum_{j\neq k} \frac{1}{(\lambda_j - m)(\lambda_j - \lambda_k)} = -\frac{1}{2}\big(T^2 - D_i\big),$$

and $\sum_j \lambda_j(\lambda_j - m)^{-1} = N - mT$, it gives $\big(\frac{5\beta}{2} - 1\big)\mathbb{E}D_i = \frac{\beta n}{2}\big(N - \mathbb{E}[mT]\big) + \frac{\beta}{2}\mathbb{E}T^2$. Eliminating $\mathbb{E}T^2$ and $\mathbb{E}[mT]$,

$$\Big(\frac{5\beta}{2} - \frac{5}{4}\Big)\mathbb{E}D_i = \frac{\beta n N}{2} - \frac{\beta n}{4}\mathbb{E}[mT] = \frac{\beta n N}{2} - \frac{n}{8}\big(\beta n \mathbb{E}m^2 - 1\big),$$

which is the identity. Dropping $\mathbb{E}m^2 \geq 0$ and using $4\beta N + 1 \leq 4\beta n$ gives the bound. □

If m concentrates under $\hat{\nu}_i$ near γ_i, the identity gives $\int D_i\, d\hat{\nu}_i \approx \frac{\beta(2\pi n \varrho(\gamma_i))^2}{10(2\beta-1)}$: the circular value, with $n/(2\pi)$ replaced by the local density $n\varrho(\gamma_i)$.

Proposition 6.4. *Let $\beta \geq 1$. For every spacing of the circular or Hermite ensemble, $a_i \leq \frac{5}{4}(\beta+1)e^{2\beta}n^{\beta+1}$. Condition 2.4 holds for every index set of the circular ensemble, and for every $I_n \subset [an, bn]$, $0 < a < b < 1$, of the Hermite ensemble.*

Proof. Fix i and $0 < x \leq 1/n$. By Chebyshev's inequality and Propositions 6.1 and 6.3, $\hat{\nu}_i(x^2 D_i > 2) \leq \frac{x^2}{2}\int D_i\, d\hat{\nu}_i \leq \frac{1}{5}$, since $\frac{2\beta}{5(2\beta-1)} \leq \frac{2}{5}$. If $x^2 D_i(y) \leq 2$, Lemma 2.2 gives $Z^{-1}\Phi_i(x,y) \geq c_i(y)e^{-\beta(1+nx^2/2)} \geq c_i(y)e^{-2\beta}$. By (8), $p_i(x) \geq \frac{4}{5}e^{-2\beta}a_i x^\beta$, and integrating over $(0, 1/n]$,

$$1 \geq \mathbb{P}(g_i \leq 1/n) \geq \frac{4e^{-2\beta}}{5(\beta+1)}\, a_i n^{-\beta-1}.$$

This is the upper bound on a_i; with Propositions 6.1 and 6.3 it gives (P3) and the upper half of (P2). In the circular case $a_i = a_\beta(n)$ does not depend on i, and $a_\beta(n) \asymp n^{\beta+1}$ by Remark 5.3. In the Hermite case Lemma 2.3 gives $a_i \geq (\beta+1)n^{\beta+1}\mathbb{P}(g_i \leq 1/n)$. By gap universality for β-ensembles with $\beta \geq 1$ [16], $n\varrho(\gamma_i)g_i$ converges in law to the Sine_β gap distribution uniformly in $i \in [an, bn]$, and this distribution charges $[0, s]$ for every $s > 0$. Since $\varrho(\gamma_i)$ is bounded on $[an, bn]$, $\mathbb{P}(g_i \leq 1/n) \geq c > 0$ there for large n. Thus $a_i \asymp n^{\beta+1}$ uniformly on I_n, which is (P1) and (P2). □

7. Proofs of Theorems A, B and C

Proof of Theorems A and B. By Proposition 6.4, Condition 2.4 holds. By Corollary 4.3, (A)–(C) hold with $\eta_n \log n \to 0$; by Proposition 2.1 the spectral gap is at least $\frac{1}{4}$; and $\Lambda_{\max}\log n \to 0$. Theorems 3.6 and 3.7 give Theorem B. In the circular case all hazards equal λ_n, hence $x_n(\tau) = \lambda_n \tau - \log n$, which is Theorem A; the three classical cases follow from Corollary 5.2. □

At $\beta = 1$ the cut-off R enters the capacity through $\log(R/\varepsilon)$, and the ratio of $\log(R/\varepsilon)$ to $\log(1/\varepsilon)$ differs from one by $\log(1/R)/\log(1/\varepsilon)$. In the slow regime $\log(1/\varepsilon)$ exceeds n, hence this ratio is $1 + O(\log n/n)$ and the logarithmic capacity costs nothing at the precision $o(1/\log n)$.

Proof of Theorem C. Write $H = \sup_n \|h_n\|$. In the proofs of Theorems 3.6 and 3.7 the initial law enters through three estimates: the sandwich (11) for separated S, the upper

bound (B) for arbitrary S, and Lemma 3.4. For separated S with $|S| \le m$, the proof of Theorem 3.7 gives $1 - c_{A_S} \le \eta_n' \to 0$ and $\lambda_S = (1 + O(\eta_n))\Lambda_S$, and by Lemma 3.5

$$\mathbb{P}_{\nu_n}(T_{A_S} > t) = \big(1 + O(\eta_n' + H\sqrt{\eta_n'})\big)e^{-\lambda_S t} + O\big(He^{-c_{\mathrm{P}} t}\big).$$

For $t \ge t_n(x_0) \ge (\log N_n + x_0)/\Lambda_{\max}$ the last term is $o(N_n^{-D})$ for every D, because $\Lambda_{\max} \log N_n \to 0$; for $S = \{i\}$ and all $t \ge 0$ the display gives $\sup_t |\mathbb{P}_{\nu_n}(\tau_i > t) - e^{-\Lambda_i t}| \to 0$. The first bound of Lemma 3.5 and (10) give (B) with an extra factor H, which multiplies terms that tend to zero in the proof of Theorem 3.6. Finally Lemma 3.5 replaces Lemma 3.4, with $\Theta_\nu = \Theta + O(H\sqrt{\eta_n'})$ and an extra factor H in the remainder. The proofs then go through verbatim. □

The example after Theorem C extends to the equilibrium measure conditioned on any event of probability bounded below, or tilted by a bounded density.

8. Variable density and the explicit centering

Proposition 8.1. *Let $0 < a < b < 1$, $I_n = \{i : an \le i \le bn\}$, and $\Lambda_i = r_n\psi(i/n)(1 + o(1/\log n))$ uniformly, where $\psi \in C^2[a,b]$ is positive and attains its minimum ψ_* only on a nonempty $E_* \subseteq \{a,b\}$, with $\psi' \ne 0$ on E_*. With $c_\psi = \sum_{e \in E_*} |\psi'(e)|^{-1}$,*

$$\psi_* r_n\, t_n(x) = \log n - \log\log n + \log(\psi_* c_\psi) + x + o(1).$$

Proof. Write $s = r_n t_n(x)$; then $s \asymp \log n$ by (12). The errors in Λ_i change $\Lambda_i t_n(x)$ by $o(1)$. Since $e^{-s\psi}$ varies on the scale $1/s \gg 1/n$, $\sum_{I_n} e^{-s\psi(i/n)} = (1 + o(1))n\int_a^b e^{-s\psi}$, and Laplace's method gives $\int_a^b e^{-s\psi} = e^{-s\psi_*} c_\psi s^{-1}(1 + O(s^{-1}))$. Thus $s\psi_* = \log n + \log c_\psi - \log s + x + o(1)$ and $\log s = \log\log n - \log\psi_* + o(1)$. □

Proposition 8.2. *Let $\beta = 2$, $I_n = \{i : \frac{n}{4} \le i \le \frac{3n}{4}\}$, $n^2\varepsilon(\log n)^{7/2} \to 0$, and assume $a_i = \frac{\pi^2}{3}(n\varrho(\gamma_i))^3(1 + o(1/\log n))$ uniformly on I_n. Then $\mathbb{P}(\Lambda_*\tau_{\mathrm{cov}} - \log n + \log\log n - \log c_* \le x) \to \exp(-e^{-x})$, with Λ_* and c_* as in the introduction.*

Proof. Here $\Lambda_i = r_n\psi(i/n)(1 + o(1/\log n))$ with $r_n = \frac{\pi^2}{3}n^2\varepsilon$ and $\psi(u) = \varrho(\gamma(u))^3$, $\gamma(u)$ the u-quantile of ϱ. Since $\gamma' = 1/\varrho(\gamma)$, $\psi' = 3\varrho\varrho'(\gamma)$; ψ is minimal exactly at $\frac14$ and $\frac34$, with $|\psi'| = 3\varrho(\gamma_*)|\varrho'(\gamma_*)|$ at both. Hence $\psi_* c_\psi = 2\varrho(\gamma_*)^2/(3|\varrho'(\gamma_*)|) = c_*$, and Theorem B with Proposition 8.1 concludes. □

The correction is absent for constant hazards, as in the circular ensemble or on mesoscopic windows of $o(n/\log n)$ indices.

9. The intermediate regime

Theorem 3.6 needs no spectral gap. In Section 4 the gap enters only through Proposition 4.2, which bounds the energy of a function vanishing on A_S by comparing it with its mean. In this section we replace the global gap by a lower bound on the conditional hazard of a spacing, and, for the Gumbel law, by a Poincaré inequality on a mesoscopic window with frozen exterior. Throughout this section $\beta > 1$.

9.1. The conditional hazard.

Lemma 9.1. *For a spacing i and fixed y, write $\ell = g_{\max}(y)$, $w(g) = Z^{-1}g^\beta\Phi_i(g,y)$ on $(0,\ell)$, $M(g) = \int_g^\ell w$, and*

$$J_i(y) = \int_\varepsilon^\ell \frac{M(g)}{w(g)}\,dg, \qquad H_i(y) = \frac{2}{\beta n\, J_i(y)}.$$

Then $\mathcal{E}_i(f) \ge \int H_i\, f^2\, d\mu$ for every $f \in \mathcal{F}_{A_i}$.

Proof. On almost every line, $u(g) = f(g, y)$ is absolutely continuous with $u(\varepsilon) = 0$, and $u(g)^2 \le \int_\varepsilon^g u'^2 w \int_\varepsilon^g w^{-1}$. Multiplying by $w(g)$ and integrating, Fubini's theorem gives $\int_\varepsilon^\ell u^2 w \le J_i(y) \int_\varepsilon^\ell u'^2 w$; we integrate over y, using $f = 0$ for $g \le \varepsilon$. □

H_i is the hazard of the i-th spacing conditionally on all other coordinates. For typical y the integral $J_i(y)$ is dominated by g near ε, where $w(g) \approx c_i(y) g^\beta$ and $M(g) \approx \int_0^\ell w$; thus $H_i(y) \approx \Lambda_i \, c_i(y)/(a_i \int_0^\ell w)$, a function with μ-mean close to Λ_i that fluctuates by a factor of order one with the positions of the neighbouring particles. The monotonicity of Φ_i gives a pointwise lower bound.

Lemma 9.2. *For every spacing i and every y, with $\ell = g_{\max}(y)$,*

$$H_i(y) \ge \frac{(\beta+1)\,\Lambda_i}{a_i\, \ell^{\beta+1}}.$$

Proof. Since $\Phi_i(\cdot, y)$ is nonincreasing on $(0, \ell)$, $w(s) \le (s/g)^\beta w(g)$ for $g < s < \ell$, and $M(g) \le w(g) g^{-\beta} \int_g^\ell s^\beta ds \le w(g)\ell^{\beta+1}/((\beta+1)g^\beta)$. Hence $J_i(y) \le \frac{\ell^{\beta+1}}{\beta+1} \int_\varepsilon^\infty g^{-\beta} dg = \frac{\ell^{\beta+1} I_\beta(\varepsilon)}{\beta+1}$, and $H_i = 2/(\beta n J_i) \ge \frac{2(\beta+1)}{\beta n \ell^{\beta+1} I_\beta(\varepsilon)} = \frac{(\beta+1)\Lambda_i}{a_i \ell^{\beta+1}}$. □

9.2. **The order of the hitting times.** Write $\delta_i = \lambda_{i+2} - \lambda_{i-1}$ (with θ in place of λ and indices mod n in the circular case). Since m lies between λ_{i-1} and λ_{i+2}, $g_{\max} \le \delta_i$.

Proposition 9.3. *Assume $a_i \le C n^{\beta+1}$. For every $L > 0$ and $t \ge 0$,*

$$\mathbb{P}(\tau_i > t) \le \exp\Big(-\frac{(\beta+1)\Lambda_i t}{C L^{\beta+1}}\Big) + \mu(n\delta_i > L).$$

Proof. Write $G = \{n\delta_i \le L\}$, an event in the coordinates y, and $c_L = (\beta+1)/(CL^{\beta+1})$. On G, Lemma 9.2 gives $H_i \ge c_L \Lambda_i$. For $f \in \mathcal{F}_{A_i}$ with $0 \le f \le 1$, since $|\nabla f|^2 \ge \langle u_i, \nabla f\rangle^2$, Lemma 9.1 gives

$$\mathcal{E}(f) \ge \mathcal{E}_i(f) \ge \int_G H_i f^2 \, d\mu \ge c_L \Lambda_i \big(\pi(f^2) - \mu(G^c)\big),$$

and Lemma 3.2 with $a = c_L \Lambda_i$, $b = c_L \Lambda_i \mu(G^c)$ concludes. □

Proof of Theorem D, parts (i) and (ii). By Proposition 6.4, Condition 2.4 holds, and $a_i \le C n^{\beta+1}$.

Lower bounds. Proposition 4.1 with $S = \{i\}$ and $nR = (n\varepsilon)^{1/2}$ gives a test function with $\pi(f^2) \ge 1 - C(n\varepsilon)^{(\beta+1)/2}$ and $\mathcal{E}(f) \le \Lambda_i/(1 - (n\varepsilon)^{(\beta-1)/2})$; by Lemma 3.1, $\mathbb{P}(\tau_i > t) \ge (1 - \eta_n) e^{-(1+\eta_n)\Lambda_i t}$ for all t, with $\eta_n \to 0$ uniformly in i. Hence $\mathbb{P}(\Lambda_i \tau_i \le s) \le 1 - (1 - \eta_n) e^{-(1+\eta_n)s}$, which is small for small s, and for $t = (1-\delta)\log|I_n|/\Lambda_{\max}$ the expected number of $i \in I_n$ with $\tau_i > t$ is at least $(1 - \eta_n)|I_n|^{1-(1+\eta_n)(1-\delta)} \to \infty$.

Tightness from above. In the circular case the spacings are exchangeable with mean $2\pi/n$, hence $\mu(n\delta_i > L) \le 6\pi/L$ by Markov's inequality. In the Hermite case, $n\varrho(\gamma_j) g_j$ converges in law uniformly in $j \in [an, bn]$ [16], so the family $n\delta_i$, $i \in I_n$, is tight. In both cases Proposition 9.3, with L large and then $t = s/\Lambda_i$ with s large, gives $\sup_{i,n} \mathbb{P}(\Lambda_i \tau_i > s) \to 0$ as $s \to \infty$. This is (i).

The cover time. By rigidity of β-ensembles in the bulk [9, 10] and of the circular β-ensemble [24], for every $\xi, D > 0$, $\mu(n\delta_i > n^\xi) \le C_{\xi,D} n^{-D}$ uniformly in $i \in I_n$. Proposition 9.3 with $L = n^\xi$ and a union bound over at most n indices give

$$\mathbb{P}\big(\Lambda_{\min}\tau_{\mathrm{cov}} > n^{\xi(\beta+2)}\big) \le n \exp(-c\, n^\xi) + C_{\xi,D} n^{1-D} \to 0.$$

For the lower bound, with i_* minimizing Λ_i, $\mathbb{P}(\Lambda_{\min}\tau_{\mathrm{cov}} < n^{-\xi}) \le \mathbb{P}(\Lambda_{i_*}\tau_{i_*} < n^{-\xi}) \to 0$ by the lower bound above. Since ξ is arbitrary, $\log(\Lambda_{\min}\tau_{\mathrm{cov}})/\log n \to 0$ in probability. This is (ii). □

Part (i) holds in the slow regime as well, where Theorems A and B give the exponential law itself. In the intermediate regime part (ii) locates τ_{cov} between $\Lambda_{\min}^{-1} n^{-o(1)}$ and $\Lambda_{\min}^{-1} n^{o(1)}$, and the expected number of uncovered spacings is still large at $(1-\delta)\log|I_n|/\Lambda_{\max}$, in agreement with the Gumbel centering $t_n(x) \asymp \log|I_n|/\Lambda$. A Gaussian tail $\mu(n\delta_i > L) \le Ce^{-cL^2}$, uniform in n, would replace $n^{o(1)}$ by a power of $\log n$.

9.3. **Windows.** Fix $K \ge 1$ and, for $i \in I_n$, the window $W_i = \{i-K, \dots, i+K+1\}$ of particles (indices mod n in the circular case). We write a configuration as (x, z) with x the window particles and z the others, μ_z for the conditional law of x given z, and

$$\mathcal{E}_z(f) = \frac{1}{\beta n}\int |\nabla_x f|^2 \, d\mu_z, \qquad \int \mathcal{E}_z(f)\,\mu(dz) \le \mathcal{E}(f).$$

For $\xi > 0$ the exterior z is *rigid* if $|(z_j - z_{j'}) - (\gamma_j - \gamma_{j'})| \le n^{-1+\xi}$ for all $j, j' \notin W_i$ with $|j-i|, |j'-i| \le 3K$, where γ_j are the classical locations ($\gamma_j = 2\pi j/n$ in the circular case). By rigidity [9, 10, 24], for every $\xi, D > 0$ the probability that the exterior of W_i is not rigid is at most $C_{\xi,D} n^{-D}$, uniformly in i when $I_n \subset [an, bn]$ in the Hermite case.

Lemma 9.4. *Let $K \ge n^{2\xi}$ and $3K < n/2$. There is C depending on a, b such that for every rigid exterior z and every f, $\mathrm{Var}_{\mu_z}(f) \le \frac{CK}{n}\mathcal{E}_z(f)$.*

Proof. The measure μ_z has density proportional to e^{-V_z} on the convex set of ordered window configurations between $z_- = z_{i-K-1}$ and $z_+ = z_{i+K+2}$, with $V_z(x) = \sum_a U(x_a) - \beta\sum_{a<b}\log|x_a - x_b|$ (with $2\sin\frac{x_a-x_b}{2}$ in the circular case) and U the confining part plus $-\beta\sum_{j\notin W_i}\log|x - z_j|$. The pair interaction is convex. For $x \in (z_-, z_+)$, rigidity gives $z_{i+K+1+k} - x \le CK/n$ for $1 \le k \le K$, hence

$$U''(x) \ge \beta\sum_{k=1}^{K}(z_{i+K+1+k} - x)^{-2} \ge \frac{\beta n^2}{C^2 K},$$

using $\frac{d^2}{du^2}(-\log|2\sin\frac{u}{2}|) = \frac{1}{4\sin^2(u/2)} \ge u^{-2}$ in the circular case. Thus $\nabla^2 V_z \ge \beta n^2/(C^2K)$, and Brascamp–Lieb gives $\mathrm{Var}_{\mu_z}(f) \le \frac{C^2K}{\beta n^2}\int|\nabla_x f|^2 d\mu_z = \frac{C^2K}{n}\mathcal{E}_z(f)$. □

Under μ_z the i-th spacing has a collision coefficient and a collision law, exactly as in Lemma 2.3: in window coordinates $e^{-V_z} = g^\beta \Phi_i^z(g, y)$, where Φ_i^z contains the interaction with the exterior and is nonincreasing in g, Lemma 2.2 holds with the same D_i, and we write $c_i^z(y) = Z_z^{-1}\Phi_i^z(0, y)$, $a_i(z) = \int c_i^z\,dy$ and $\hat\nu_i^z = c_i^z\,dy/a_i(z)$. Since $p_i = \int p_i^z\,\mu(dz)$ and $x^{-\beta}p_i^z(x) \le a_i(z)$ increases to $a_i(z)$ as in Lemma 2.3,

$$\int a_i(z)\,\mu(dz) = a_i, \qquad \int a_i(z)\,\hat\nu_i^z(D_i > Kn^2)\,\mu(dz) = a_i\,\hat\nu_i(D_i > Kn^2) \le \frac{C_0 a_i}{K}. \tag{14}$$

Hypothesis 9.5. There are $\xi > 0$ and η_n with $\eta_n \log n \to 0$ such that, with $K = n^{2\xi}$, for all $i \in I_n$ and every rigid exterior z of W_i,

(Q1) $a_i(z) \ge (1-\eta_n)\,a_i$;
(Q2) $\hat\nu_i^z(D_i > Kn^2) \le \eta_n$.

By (14) both conditions hold on average over the exterior; Hypothesis 9.5 asks for them uniformly over rigid exteriors. It is a statement on local equilibrium measures alone: a one-sided local law for their collision coefficients, and a moment bound for their collision laws. It controls the conditional hazard as follows.

Lemma 9.6. *Let $\beta > 1$, assume Condition 2.4 and Hypothesis 9.5, and suppose $\theta_K := K^{2\beta+1}(n\varepsilon)^{\beta-1} \to 0$. Then, with $M = CK^{(\beta+1)/2}$, for all $i \in I_n$ and every rigid exterior z,*

$$\int \min(H_i, M\Lambda_i)\,d\mu_z \ge (1 - C\theta_K - C(n\varepsilon\sqrt{K})^{\min(\beta-1,1)})(1-\eta_n)^2\,\Lambda_i. \tag{15}$$

Proof. We work under μ_z, with $w(g) = Z_z^{-1} g^\beta \Phi_i^z(g,y)$, $m(y) = \int_0^\ell w$ the density of the y-marginal, and $\ell = g_{\max}(y)$. All window particles lie between z_- and z_+, and rigidity gives $\ell \le z_+ - z_- \le CK/n$. Write $r_0 = (Kn^2)^{-1/2}$ and $G = \{D_i \le Kn^2\}$. For $y \in G$ and $g \le r_0$, $g^2 D_i \le 1$, and Lemma 2.2 gives $w(g) \ge c_i^z(y) g^\beta e^{-x(g)}$ with $x(g) = \frac{\beta}{2} g^2 (D_i + n) \le \beta$. Hence $m(y) \ge e^{-\beta} c_i^z(y) r_0^{\beta+1}/(\beta+1)$, that is,

$$c_i^z(y)/m(y) \le C(Kn^2)^{(\beta+1)/2}.$$

We bound $J_i(y) = \int_\varepsilon^\ell M(g) w(g)^{-1} dg$ on G. On $[\varepsilon, r_0]$, $M \le m$ and the computation in the proof of Proposition 4.2, with R replaced by r_0, gives $\int_\varepsilon^{r_0} M/w \le (m/c_i^z) I_\beta(\varepsilon)(1 + C(\varepsilon/r_0)^{\min(\beta-1,1)})$, and $\varepsilon/r_0 = n\varepsilon\sqrt{K}$. On $[r_0, \ell]$, the argument of Lemma 9.2 gives $M/w \le \ell^{\beta+1}/((\beta+1) g^\beta)$, hence

$$\int_{r_0}^\ell \frac{M}{w} \le \frac{\ell^{\beta+1} r_0^{1-\beta}}{\beta^2 - 1} = \frac{m}{c_i^z} I_\beta(\varepsilon) \cdot \frac{c_i^z}{m} \frac{\ell^{\beta+1}}{\beta+1} \Big(\frac{\varepsilon}{r_0}\Big)^{\beta-1} \le \frac{m}{c_i^z} I_\beta(\varepsilon) \cdot C K^{\frac{3(\beta+1)}{2}} K^{\frac{\beta-1}{2}} (n\varepsilon)^{\beta-1}.$$

The last factor is $C\theta_K$. Thus, on G,

$$J_i \le \frac{m}{c_i^z} I_\beta(\varepsilon)(1 + \theta'), \qquad \theta' = C\theta_K + C(n\varepsilon\sqrt{K})^{\min(\beta-1,1)},$$

and

$$H_i = \frac{2}{\beta n J_i} \ge (1 - \theta') \frac{\Lambda_i}{a_i} \frac{c_i^z}{m} \qquad \text{on } G.$$

By (P2) and the bound on c_i^z/m, $\frac{\Lambda_i}{a_i} \frac{c_i^z}{m} \le M\Lambda_i$ for $M = CK^{(\beta+1)/2}$, so the truncation is inactive on G. Integrating against $m(y)\,dy$,

$$\int \min(H_i, M\Lambda_i)\, d\mu_z \ge (1 - \theta') \frac{\Lambda_i}{a_i} \int_G c_i^z \, dy = (1 - \theta')\, \Lambda_i \, \frac{a_i(z)}{a_i} \, \hat{\nu}_i^z(G),$$

and (Q1), (Q2) conclude. □

Remark 9.7. Condition (Q1) is a comparison of small-gap probabilities. By Lemma 2.3 under μ_z, $a_i(z) \ge (\beta+1) x^{-\beta-1} \mu_z(g_i \le x)$; by the proof of Proposition 6.4, $\mathbb{P}(g_i \le x) \ge \frac{a_i x^{\beta+1}}{\beta+1}(1 - C(nx)^2)$. With $x = s/n$ and $s = (\log n)^{-1}$, (Q1) follows from $\mu_z(g_i \le s/n) \ge (1 - o(1/\log n))\, \mu(g_i \le s/n)$, a lower bound by a quantity of order $s^{\beta+1}$, polylogarithmic in n. In the Hermite case, the comparison theorem of Erdős and Yau for local equilibrium measures [16, Thm. 4.1] bounds the difference of single-gap statistics under two boundary conditions satisfying its hypotheses by $CK^{-c}\|O'\|_\infty$, also for different external potentials, and an affine change of variables matching the configuration intervals of two rigid exteriors changes the potential and rescales the gaps by a factor $1 + O(n^\xi/K)$; smoothing the indicator at relative scale $(\log n)^{-2}$ costs a polylogarithmic factor, which K^{-c} absorbs. Thus (Q1) holds for every exterior satisfying the hypotheses (4.7)–(4.10) of [16], as soon as these exteriors carry μ-probability $1 - o((\log n)^{-\beta-2})$. Two points remain: that set must have probability $1 - O(n^{-D})$ to be used in Proposition 9.8, and (Q2) is a moment bound for the collision law of a local equilibrium measure, the local analogue of Proposition 6.3.

Proposition 9.8. *Assume (P1), rigidity as above, and* $\int \min(H_i, M\Lambda_i)\, d\mu_z \ge (1 - \eta_n)\Lambda_i$ *for all* $i \in I_n$ *and every rigid exterior* z. *For every* D *there is* C *such that for every* $S \subset I_n$ *with* $|S| \le m$ *and every* $f \in \mathcal{F}_{A_S}$ *with* $0 \le f \le 1$,

$$\mathcal{E}(f) \ge \Big(1 - \eta_n - CM\Big(\frac{K\Lambda_S}{n}\Big)^{1/2} - CM\frac{K\Lambda_S}{n}\Big) \Lambda_S^{\mathrm{eff}}\, \pi(f^2) - C\Lambda_S n^{-D},$$

with $\Lambda_S^{\mathrm{eff}} = \sum_C \mu_C^{-1} \sum_{i \in C} \Lambda_i$ *as in Proposition 4.2.*

Proof. If $\mathcal{E}(f) > 2\Lambda_S \pi(f^2)$ there is nothing to prove, and we assume the contrary. By (13) and Lemma 9.1 it suffices to show, for $i \in S$,

$$\int H_i f^2 \, d\mu \geq \Big(1 - \eta_n - CM\Big(\frac{K\Lambda_S}{n}\Big)^{1/2} - CM\frac{K\Lambda_S}{n}\Big)\Lambda_i \pi(f^2) - C\Lambda_i n^{-D}.$$

Write $h = \min(H_i, M\Lambda_i) \leq H_i$ and $\int h f^2 \, d\mu = \int \mu(dz)\big[\mu_z(h)\mu_z(f^2) + \mathrm{Cov}_{\mu_z}(h, f^2)\big]$. We drop the non-negative contribution of non-rigid exteriors. For rigid z, $\mu_z(h) \geq (1-\eta_n)\Lambda_i$ by assumption, and, since $0 \leq h \leq M\Lambda_i$ and $0 \leq f \leq 1$,

$$|\mathrm{Cov}_{\mu_z}(h, f^2)| \leq M\Lambda_i \|f^2 - \mu_z(f^2)\|_{L^1(\mu_z)} \leq M\Lambda_i\big(2\|f - \mu_z(f)\|_{L^2(\mu_z)}\|f\|_{L^2(\mu_z)} + \mathrm{Var}_{\mu_z}(f)\big).$$

By Lemma 9.4, $\mathrm{Var}_{\mu_z}(f) \leq CK\mathcal{E}_z(f)/n$. Integrating over z with the Cauchy–Schwarz inequality,

$$\int \|f - \mu_z(f)\|_{L^2(\mu_z)}\|f\|_{L^2(\mu_z)} \, \mu(dz) \leq (CK\mathcal{E}(f)/n)^{1/2}\pi(f^2)^{1/2},$$

and $\int_{\text{rigid}} \mu_z(f^2)\,\mu(dz) \geq \pi(f^2) - C_D n^{-D}$ because $f^2 \leq 1$. With $\mathcal{E}(f) \leq 2\Lambda_S\pi(f^2)$ this is the claim. □

Proof of Theorem D, part (iii). By Proposition 6.4, Condition 2.4 holds. Since $2\xi(2\beta+1) < \delta(\beta-1)$, we have $K^{2\beta+1}(n\varepsilon)^{\beta-1} \leq n^{-c}$, and Lemma 9.6 gives the assumption of Proposition 9.8 with $M = CK^{(\beta+1)/2}$ and an error $\eta_n + n^{-c}$; moreover $M^2 K\Lambda_{\max}/n \leq CK^{\beta+2}(n\varepsilon)^{\beta-1} \leq n^{-c}$. Condition (A) of Section 3 is (P1). For (B) we apply Lemma 3.2 to A_S with the inequality of Proposition 9.8: $a = (1-\eta_n')\Lambda_S^{\mathrm{eff}}$ with $\eta_n' = \eta_n + CM(K\Lambda_{\max}/n)^{1/2}$, and $b = C\Lambda_S n^{-D}$, so that $b/a \leq Cn^{-D}$; (9) follows from $\Lambda_S^{\mathrm{eff}} \geq \sum_C w_C$, and $\eta_n' \log n \to 0$. For (C), Proposition 4.1 with $nR = (\log n)^{-1}$ gives test functions with $\eta_n \leq C((\log n)^{-\beta-1} + (n\varepsilon \log n)^{\beta-1})$, and $\eta_n \log n \to 0$ because $n\varepsilon \leq n^{-\delta}$. Theorem 3.6 concludes. □

Part (iii) covers every ε with $n\varepsilon \leq n^{-\delta}$, for any $\delta > 0$: all of the intermediate regime up to its upper end. The infinite-volume Dyson Brownian motion has no spectral gap [30]; the window inequality of Lemma 9.4 replaces it, at the price of the static Hypothesis 9.5.

10. Near-degeneracies of noisy quantum evolutions

Let H_t be a Hermitian Brownian motion on $n \times n$ matrices, with independent entries above the diagonal and $\mathbb{E}|dH_{jk}|^2 = dt/n$ for all j, k, and let U_t solve

$$dU_t = i\, dH_t\, U_t - \tfrac{1}{2} U_t\, dt, \qquad U_0 \text{ Haar distributed.}$$

Then U_t is a Brownian motion on the unitary group, stationary under Haar measure, and its eigenvalues $e^{i\theta_k(t)}$ solve (4) with $\beta = 2$ [15]: second-order perturbation theory gives the drift $\frac{1}{2}\sum_{j\neq k}\mathbb{E}|dH_{jk}|^2 \cot\frac{\theta_k - \theta_j}{2}$, and the diagonal entries give the noise. Physically, U_t is the evolution operator of an n-level quantum system driven by white noise, the model of the Brownian circuits used to study fast scrambling [25], and the θ_k are its quasienergies. A near-degeneracy $|\theta_{k+1} - \theta_k| \leq \varepsilon$ is an avoided crossing of width ε. When the noise is replaced by a slow deterministic drive, Landau–Zener transitions between neighbouring eigenstates take place at such crossings.

Corollary 10.1. *In this setting let $\varepsilon_n n^2 (\log n)^{7/2} \to 0$ and $T_n = 24\pi/((n^2-1)\varepsilon_n)$. Then:*

- *(i) the time at which a given pair of neighbouring quasienergies first comes within ε_n is asymptotically exponential with mean T_n;*
- *(ii) the time at which every pair of neighbouring quasienergies has done so equals $T_n(\log n + G_n)$, with G_n converging in law to the standard Gumbel distribution;*
- *(iii) the number of pairs that have not come within ε_n by time $T_n(\log n + x)$ converges in law to a Poisson variable with mean e^{-x}, and the late hitting times, in the scale $t/T_n - \log n$, converge to a Poisson process with intensity $e^{-x}dx$.*

Proof. By the above, the quasienergies form the circular Dyson process at $\beta = 2$ in equilibrium, and Theorem A applies with $\lambda_n = 1/T_n$. □

The Brownian motions on the symmetric spaces $U(n)/O(n)$ and $U(2n)/Sp(n)$ give the cases $\beta = 1$ and $\beta = 4$ [15], systems with time-reversal symmetry, with or without Kramers degeneracy. With the normalization of (4), Theorem A gives the rates $(n^2 - 1)/(12n \log(1/\varepsilon))$ and $(n^2 - 1)(4n^2 - 1)\varepsilon^3/(720\pi)$ per pair. With time-reversal symmetry the rate depends on ε only through $\log(1/\varepsilon)$: near-degeneracies of every width occur on a time scale that grows logarithmically in $1/\varepsilon$.

The exponents have a geometric origin. Near a degeneracy of two neighbouring levels the gap is the norm of a vector in $\mathbb{R}^{\beta+1}$: the difference of the two diagonal entries of the 2×2 block and the β real components of its off-diagonal entry. Degeneracies therefore have codimension $\beta + 1$, as von Neumann and Wigner observed [31]. Under white-noise driving this vector performs a $(\beta + 1)$-dimensional Brownian motion, whose norm is the Bessel process of Section 1. A Brownian path reaches the ε-neighbourhood of a set of codimension d at a rate proportional to the Newtonian capacity of a d-dimensional ball of radius ε, that is ε^{d-2} for $d \geq 3$ and $1/\log(1/\varepsilon)$ for $d = 2$; this is the factor $1/I_\beta(\varepsilon)$ in (3). A smooth one-parameter family of Hamiltonians, on the other hand, passes within ε of a set of codimension d with probability proportional to ε^{d-1}, which is the classical law Δ^β for the number of avoided crossings of gap below Δ per unit parameter [33, 34]. Noise lowers the exponent by one, because rough paths explore the transverse directions of the degeneracy set.

11. Concluding remarks

11.1. **Adjacent spacings.** Clusters of adjacent spacings are removed in Theorem 3.6 by the factor $\frac{4}{3}$ of (10), which is far from sharp. Statically, two adjacent spacings are both below s with probability of order $s^{3\beta+2}$, not $s^{2\beta+2}$; at $\beta = 2$ the constant is explicit.

Proposition 11.1. *For the* Sine_2 *process with unit density, and* g_1, g_2 *the two spacings following the point at the origin under the Palm measure,*

$$\mathbb{P}(g_1 \leq s,\, g_2 \leq s) \sim \frac{31\pi^6}{16200}\, s^8, \qquad s \downarrow 0.$$

Proof. Write $K(x, y) = \frac{\sin \pi(x-y)}{\pi(x-y)}$. The Palm measure at 0 is determinantal with kernel $K_0(x, y) = K(x, y) - K(x, 0)K(0, y)$ [29]. Its two-point function $\rho(a, b) = K_0(a, a)K_0(b, b) - K_0(a, b)^2$ vanishes to second order on $\{a = 0\}$, $\{b = 0\}$ and $\{a = b\}$, and expanding the sine kernel at the origin gives $\rho(a, b) = \frac{\pi^6}{135} a^2 b^2 (a-b)^2 + O(|(a, b)|^8)$. By inclusion–exclusion for determinantal processes, $\mathbb{P}(g_1 \leq s, g_2 \leq s)$ equals the integral of ρ over $\{0 < a < s,\ a < b < a + s\}$ up to terms of higher order, and $\int_0^s \int_0^s a^2 u^2 (a + u)^2 \, du \, da = \frac{31}{120} s^8$. □

11.2. **The collision coefficient of the Gaussian unitary ensemble.** Proposition 8.2 assumes $a_i = \frac{\pi^2}{3}(n\varrho(\gamma_i))^3(1 + o(1/\log n))$. Proposition 6.4 gives $a_i \asymp n^3$ on bulk windows, and the constant holds on average. Indeed, as in the proof of Proposition 5.1, for a Borel set $B \subset (-2, 2)$,

$$\sum_i a_i \, \hat{\nu}_i(m \in B) = \int_B \kappa_n(u) \, du, \qquad \kappa_n(u) = K_n(u, u)\, \partial_1 \partial_2 K_n(u, u) - \big(\partial_2 K_n(u, u)\big)^2,$$

with K_n the Hermite kernel, since $\rho_2(u, u + x) = \kappa_n(u) x^2 + O(x^3)$ and pairs at distance x are consecutive up to $O(x^5)$. In the bulk, $K_n(u + \frac{a}{n\varrho}, u + \frac{b}{n\varrho})/(n\varrho)$ converges with its derivatives to the sine kernel $s(a - b)$ [4, Ch. 3], and $-s''(0) = \pi^2/3$, $s'(0) = 0$; hence $\kappa_n(u) = \frac{\pi^2}{3}(n\varrho(u))^4(1 + o(1))$ locally uniformly in $(-2, 2)$. Since the indices i with $\gamma_i \in B$ have density $n\varrho$, this is the assumption of Proposition 8.2 averaged over the indices whose

collisions fall in B. The pointwise statement needs that a collision of the i-th spacing occurs within $o(n/\log n)$ indices of γ_i under $\hat{\nu}_i$, and an error $o(1/\log n)$ in the bulk kernel asymptotics.

11.3. **Questions.** In the slow regime the picture is complete for the circular ensemble at every $\beta \geq 1$ and on bulk windows of the Hermite ensemble: the late near-collisions form a Poisson process with intensity $e^{-x}dx$ in the variable $x_n(\tau)$, the cover time has Gumbel fluctuations, and each hitting time is exponential with rate Λ_i, as soon as $\Lambda_i(\log n)^{(\beta+5)/2} \to 0$. The rate is the flux of a Bessel process of dimension $\beta+1$ weighted by the collision coefficient, and it is exact because the configuration seen from a collision has an explicitly computable inverse square moment.

In the intermediate regime Theorem D determines the order of each hitting time and of the cover time up to $n^{o(1)}$. The Gumbel law there rests on Hypothesis 9.5. By Remark 9.7, its first half reduces to the comparison theorem of Erdős and Yau for local equilibrium measures, up to the probability of their boundary conditions; its second half is a local version of the loop identity of Proposition 6.3, in which the exterior enters through the external force on the window. The point process in that regime further needs multi-time survival bounds without a global gap. For the explicit centering in the Hermite ensemble the collision coefficients are needed index by index to the same precision. For initial data far from equilibrium, such as a deterministic configuration or the spectrum of a Wigner matrix, the density h is not in $L^2(\mu)$ or has norm $e^{O(n^2)}$; the Markov property at a time s of order $\log \|h\|$ reduces the problem to Theorem C, and it remains to show that near-collisions before time s do not correlate with late survival. For $\beta < 1$ the spacings collide, the hitting time of $\{g_i \leq \varepsilon\}$ stays of order one as $\varepsilon \downarrow 0$, and near-collisions are no longer rare events.

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Universidad Panamericana, Mexico City, Mexico
Email address: jcgalindo@up.edu.mx